

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))

    df = df[cols]
```

```
#df['campaign'] = "ALL"
```

```
return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/switched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	7988
Nombre de lignes df clean	7988
Nombre de features	77
% de NaN	0.0
Taille dataset entraînement	5591
Taille dataset test	2396
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_NH3_st_24h_mean, r_NH3_st_24h_std, r_NH3_st_24h_rms, r_NH3_st_24h_median, r_NH3_st_24h_min, r_NH3_st_24h_max, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_temperature_st_24h_mean, r_temperature_st_24h_std, r_temperature_st_24h_rms, r_temperature_st_24h_median, r_temperature_st_24h_min, r_temperature_st_24h_max, r_temperature_st_24h_diff_last, r_temperature_st_24h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_12h_mean, r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_24h_mean, r_umidity_st_24h_std, r_umidity_st_24h_rms, r_umidity_st_24h_median, r_umidity_st_24h_min, r_umidity_st_24h_max, r_umidity_st_24h_diff_last, r_umidity_st_24h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV

- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

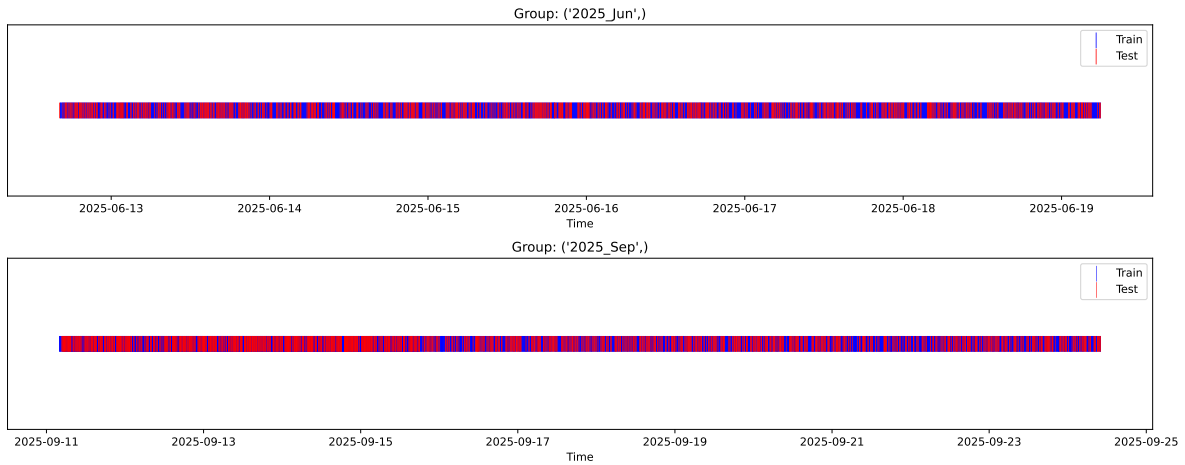
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

External Split: Train: 5591 samples Test: 2397 samples



2.2.2 Internal CV Folds

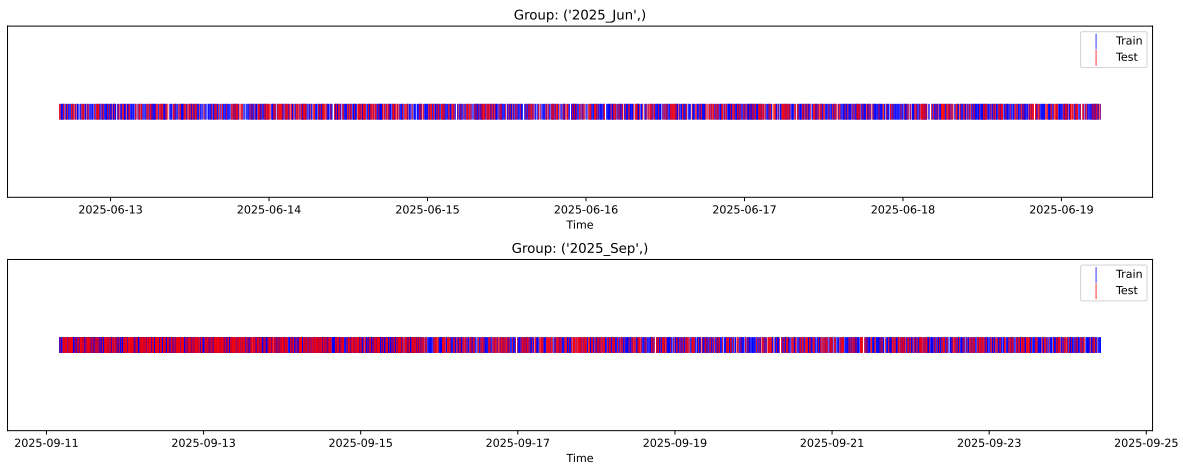
Fold 1/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



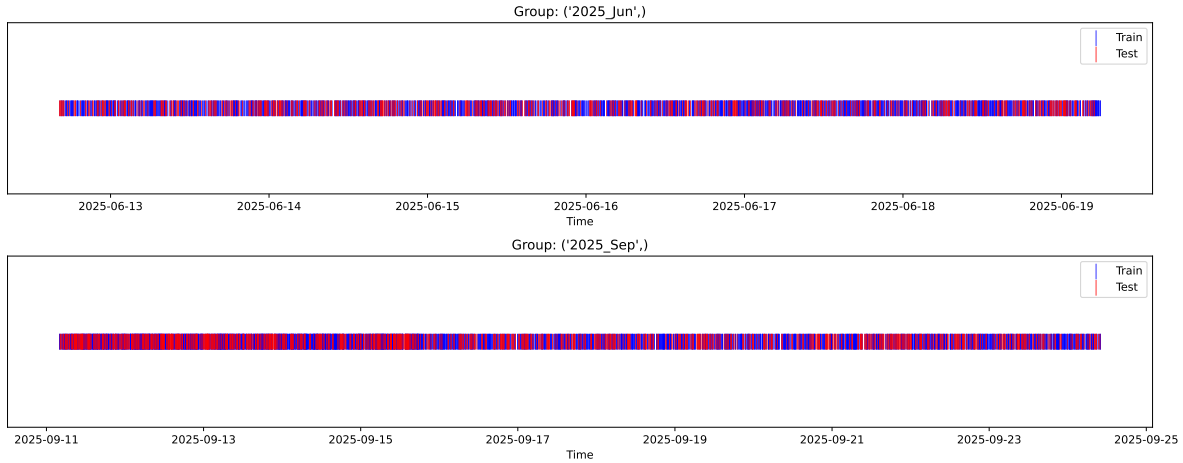
Fold 4/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.350, std=0.793, min=0.959, max=8.682
y_test: mean=2.359, std=0.815, min=0.973, max=13.845

Fold 0 — y_test: mean=2.36, std=0.79, y_train: mean=2.35, std=0.79

Fold 1 — y_test: mean=2.35, std=0.78, y_train: mean=2.35, std=0.80

Fold 2 — y_test: mean=2.34, std=0.79, y_train: mean=2.35, std=0.79

Fold 3 — y_test: mean=2.37, std=0.83, y_train: mean=2.34, std=0.77

Fold 4 — y_test: mean=2.37, std=0.79, y_train: mean=2.34, std=0.79

train scores CV: [0.58890142 0.57746942 0.58050784 0.58638309 0.58768626]

test scores CV: [0.5543343 0.57760276 0.57313542 0.55795676 0.5583262]

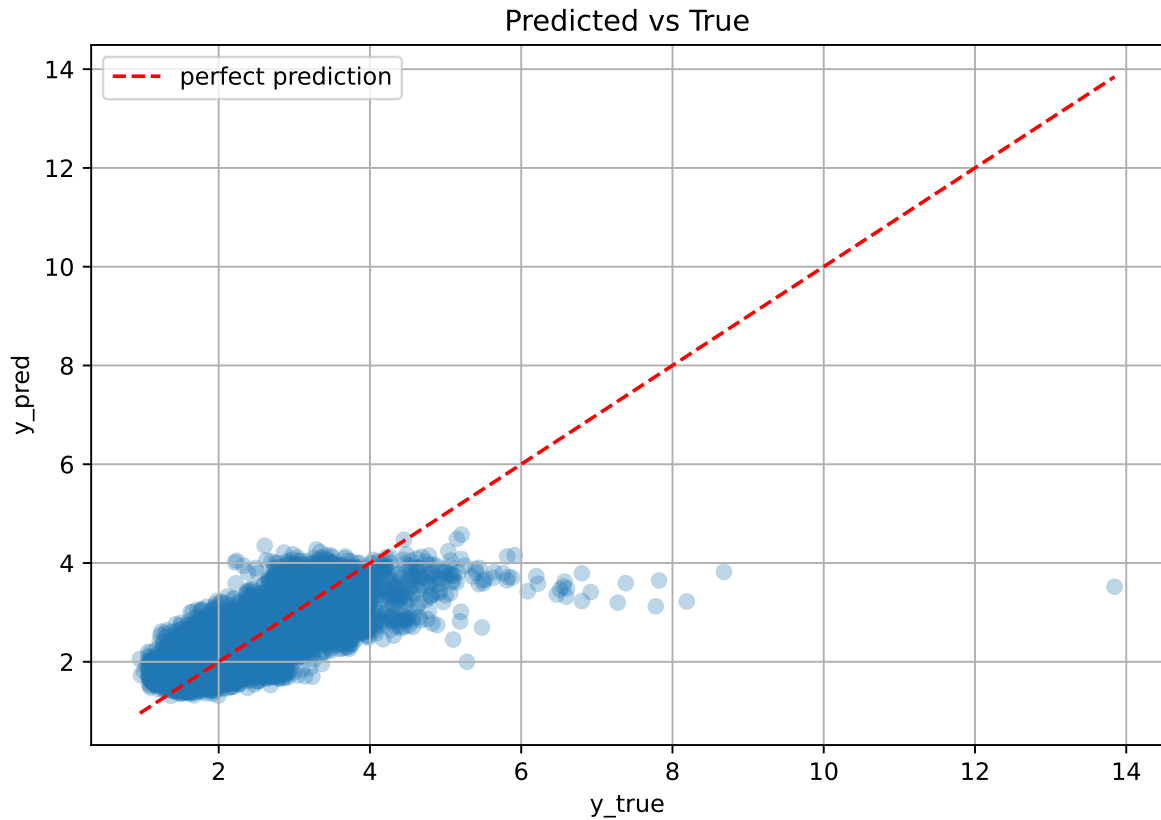
rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5842	0.5643	0.0093	0.5412	0.5451	-0.0191	0.9661	1.0353

Coefficients:

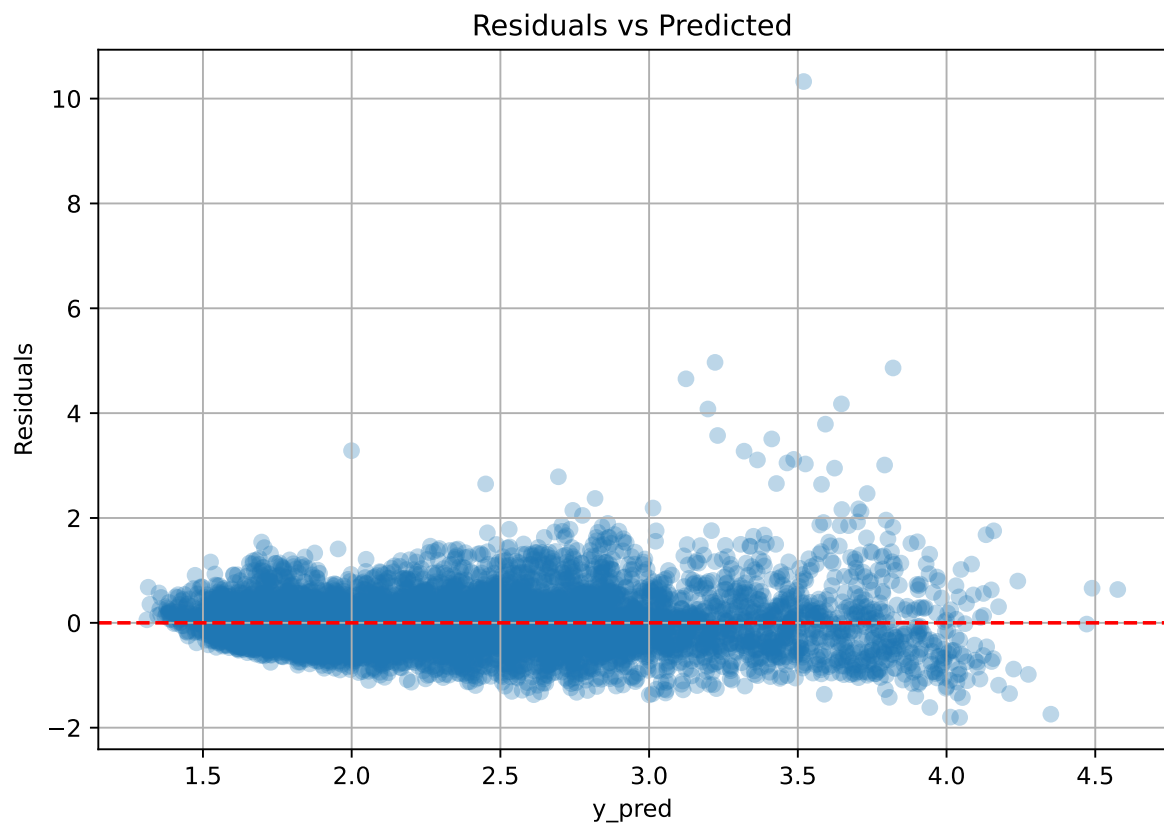
coef_r_CO2 = -0.0455, coef_r_NH3 = 0.4641, coef_r_NH3_st_12h_diff_last = 0.0005, coef_r_NH3_st_12h_max = 0.2066, coef_r_NH3_st_12h_mean = -10.0977, coef_r_NH3_st_12h_median = 0.1194, coef_r_NH3_st_12h_min = -0.4552, coef_r_NH3_st_12h_rms = 8.7337, coef_r_NH3_st_12h_slope = -0.0557, coef_r_NH3_st_12h_std = -0.0621, coef_r_NH3_st_24h_diff_last = 0.0005, coef_r_NH3_st_24h_max = 0.3763, coef_r_NH3_st_24h_mean = -2.1380, coef_r_NH3_st_24h_median = -0.1632, coef_r_NH3_st_24h_min = -0.5207, coef_r_NH3_st_24h_rms = 3.7320, coef_r_NH3_st_24h_slope = 0.1751, coef_r_NH3_st_24h_std = -0.0654, coef_r_NH3_st_8h_diff_last = 0.0005, coef_r_NH3_st_8h_max = -1.7669, coef_r_NH3_st_8h_mean = 8.1154, coef_r_NH3_st_8h_median = 0.7600, coef_r_NH3_st_8h_min = 1.1808, coef_r_NH3_st_8h_rms = -8.5691, coef_r_NH3_st_8h_slope = 0.0124, coef_r_NH3_st_8h_std = 0.1991, coef_r_THI = 0.3688, coef_r_temperature = -0.0208, coef_r_temperature_st_12h_diff_last = 0.0086, coef_r_temperature_st_12h_max = -0.4774, coef_r_temperature_st_12h_mean = -3.5842, coef_r_temperature_st_12h_median = -0.7991, coef_r_temperature_st_12h_min = -1.5625, coef_r_temperature_st_12h_rms = 11.7541, coef_r_temperature_st_12h_slope = 0.8010, coef_r_temperature_st_12h_std = -0.0751, coef_r_temperature_st_24h_diff_last = 0.0086, coef_r_temperature_st_24h_max = 0.0034, coef_r_temperature_st_24h_mean = 1.6538, coef_r_temperature_st_24h_median = 1.5718, coef_r_temperature_st_24h_min = 0.3310, coef_r_temperature_st_24h_rms = -6.1672, coef_r_temperature_st_24h_slope = -1.1522, coef_r_temperature_st_24h_std = 0.2219, coef_r_temperature_st_8h_diff_last = 0.0086, coef_r_temperature_st_8h_max = 0.2051, coef_r_temperature_st_8h_mean = 67.8624, coef_r_temperature_st_8h_median = -1.0781, coef_r_temperature_st_8h_min = -0.0633, coef_r_temperature_st_8h_rms = -69.8581, coef_r_temperature_st_8h_slope = -0.1309, coef_r_temperature_st_8h_std = 0.0461, coef_r_umidity = -0.8469, coef_r_umidity_st_12h_diff_last = -0.0021, coef_r_umidity_st_12h_max = 0.1679, coef_r_umidity_st_12h_mean = -5.6669, coef_r_umidity_st_12h_median = 0.0911,

coef_r_umidity_st_12h_min = -0.7543, coef_r_umidity_st_12h_rms = 4.6015,
coef_r_umidity_st_12h_slope = -0.3897, coef_r_umidity_st_12h_std = -0.2440,
coef_r_umidity_st_24h_diff_last = -0.0021, coef_r_umidity_st_24h_max = -
0.1622, coef_r_umidity_st_24h_mean = 2.3700, coef_r_umidity_st_24h_median
= 0.2771, coef_r_umidity_st_24h_min = 0.0173, coef_r_umidity_st_24h_rms =
-3.5094, coef_r_umidity_st_24h_slope = -0.3883, coef_r_umidity_st_24h_std =
0.0241, coef_r_umidity_st_8h_diff_last = -0.0021, coef_r_umidity_st_8h_max =
0.3680, coef_r_umidity_st_8h_mean = 25.8554, coef_r_umidity_st_8h_median =
0.3124, coef_r_umidity_st_8h_min = 0.4159, coef_r_umidity_st_8h_rms = -23.8587,
coef_r_umidity_st_8h_slope = 0.2033, coef_r_umidity_st_8h_std = 0.1928, intercept =
2.3500,

2.3.2 Scatter



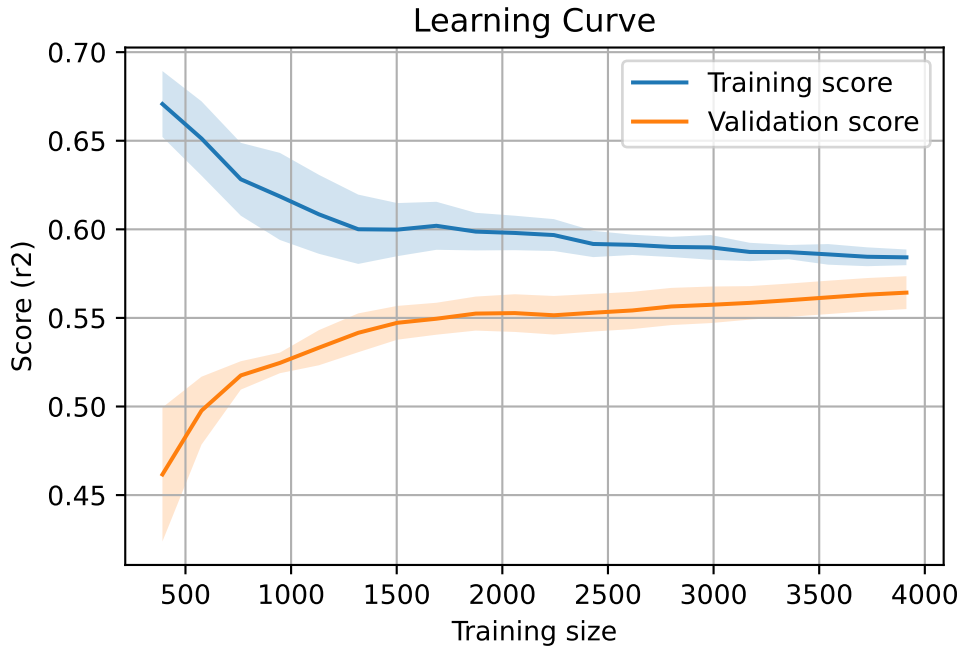
2.3.3 Scatter residuals



2.3.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396



2.4 Model : LGBM

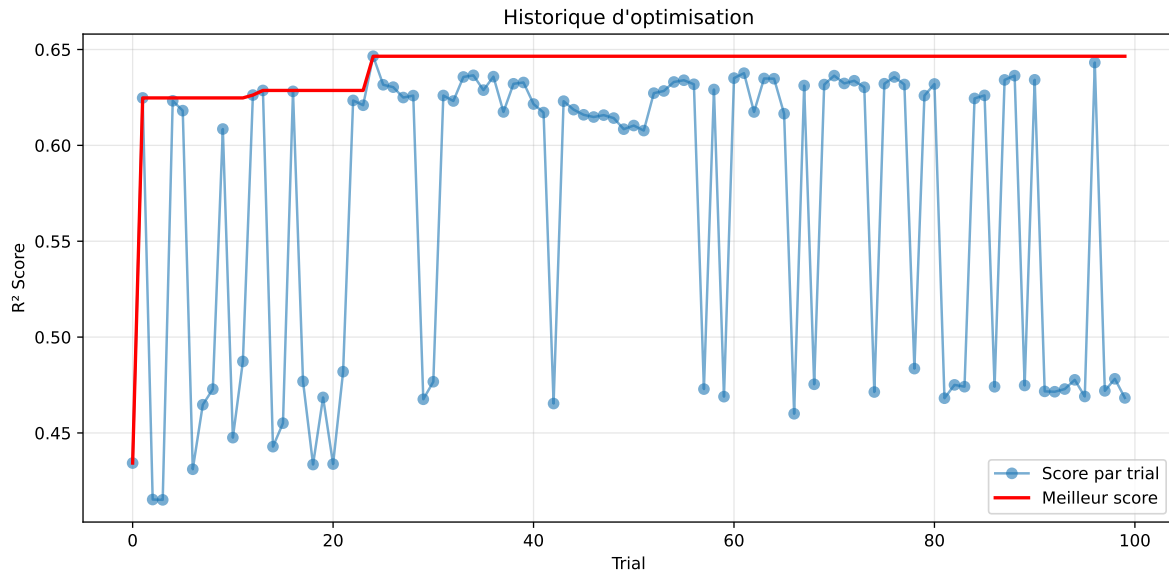
2.4.1 Gridsearch

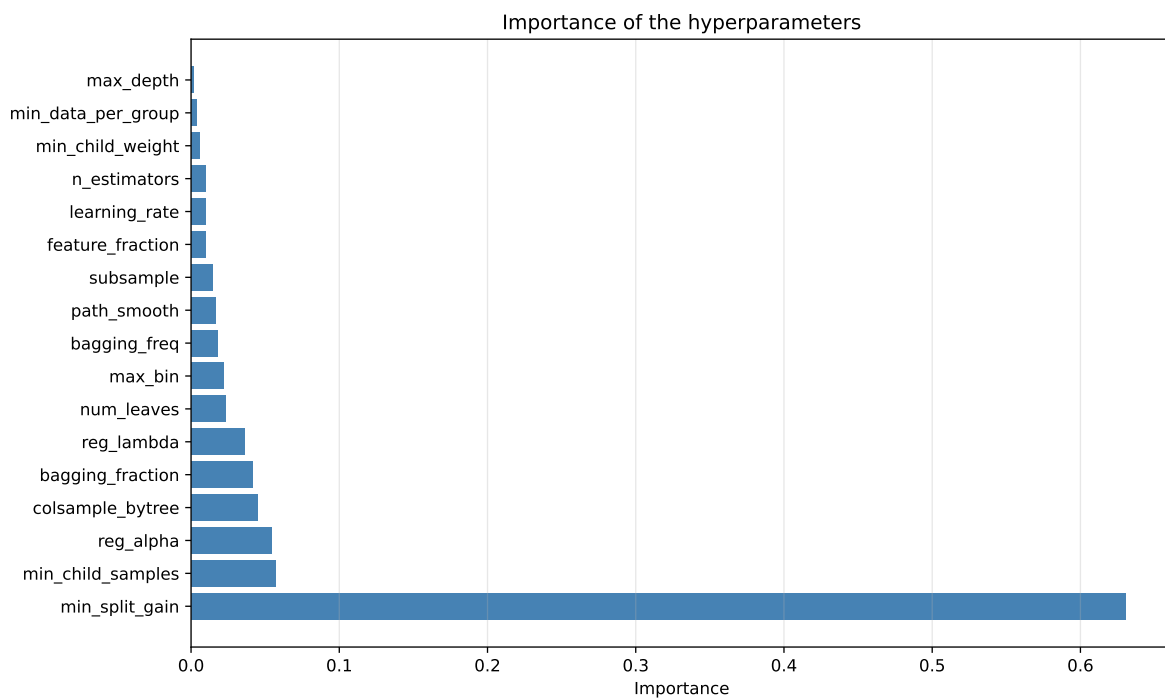
Distribution y_train vs y_test: y_train: mean=2.350, std=0.793, min=0.959, max=8.682
y_test: mean=2.359, std=0.815, min=0.973, max=13.845

===== TOP Importance FEATURES =====

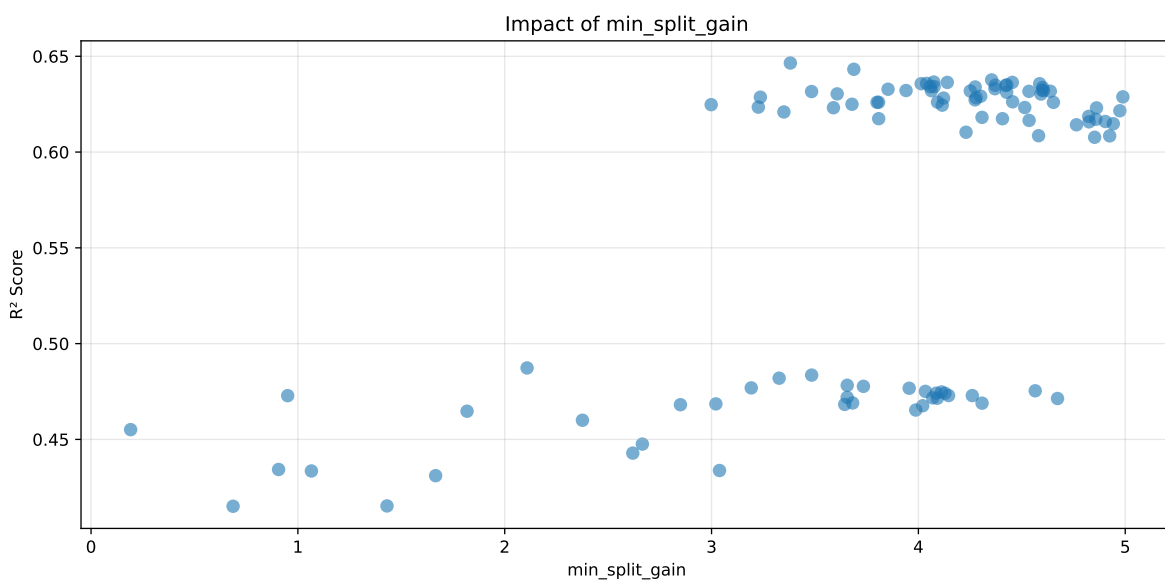
feature	importance
r_CO2	110
r_umidity	56
r_NH3_st_24h_min	41
r_temperature_st_12h_slope	35
r_NH3_st_8h_std	31
r_temperature_st_24h_max	26
r_temperature_st_8h_slope	24
r_NH3	22
r_temperature_st_8h_diff_last	22
r_temperature_st_24h_slope	16
r_umidity_st_8h_diff_last	15
r_NH3_st_24h_slope	15
r_NH3_st_24h_max	15
r_umidity_st_8h_min	13
r_NH3_st_24h_std	11
r_temperature_st_12h_std	11
r_NH3_st_8h_diff_last	11
r_NH3_st_12h_std	10
r_umidity_st_12h_min	9
r_NH3_st_24h_rms	9
r_NH3_st_8h_slope	8
r_temperature_st_24h_median	8
r_temperature_st_24h_std	7
r_umidity_st_12h_slope	7
r_umidity_st_24h_max	7
r_umidity_st_24h_min	5
r_umidity_st_24h_slope	5
r_NH3_st_12h_slope	5
r_NH3_st_24h_median	5
r_umidity_st_12h_std	4
r_temperature_st_12h_diff_last	4
r_umidity_st_8h_slope	4
r_temperature_st_12h_min	4
r_temperature_st_8h_min	4
r_umidity_st_24h_mean	3
r_umidity_st_8h_median	3
r_umidity_st_12h_max	3
r_temperature_st_8h_std	3
r_NH3_st_24h_mean	3
r_umidity_st_8h_max	2
r_umidity_st_8h_mean	2
r_NH3_st_12h_mean	2
r_umidity_st_12h_diff_last	2

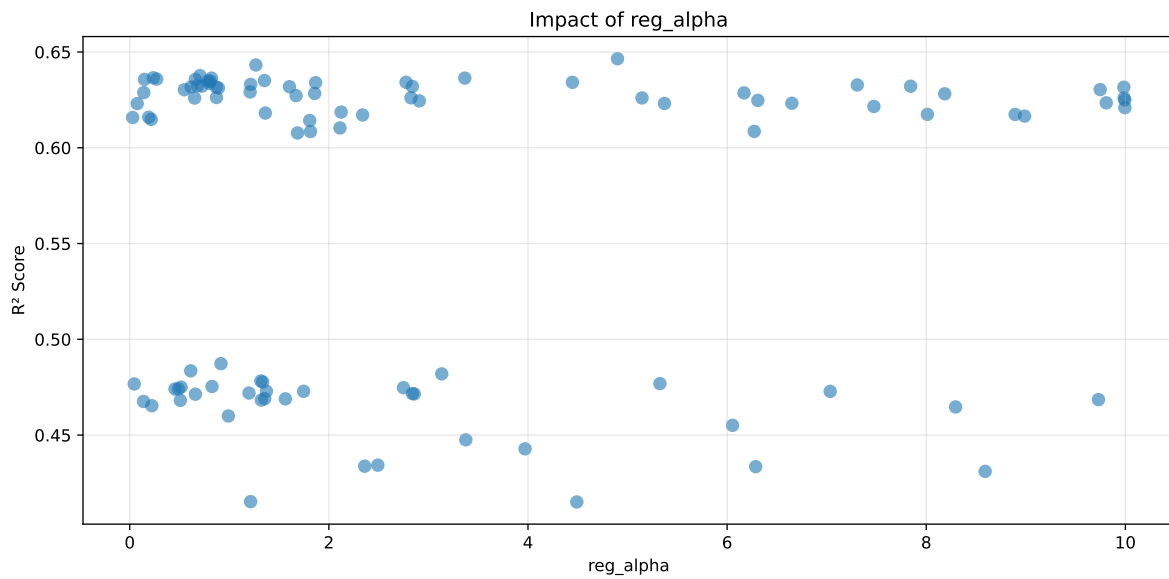
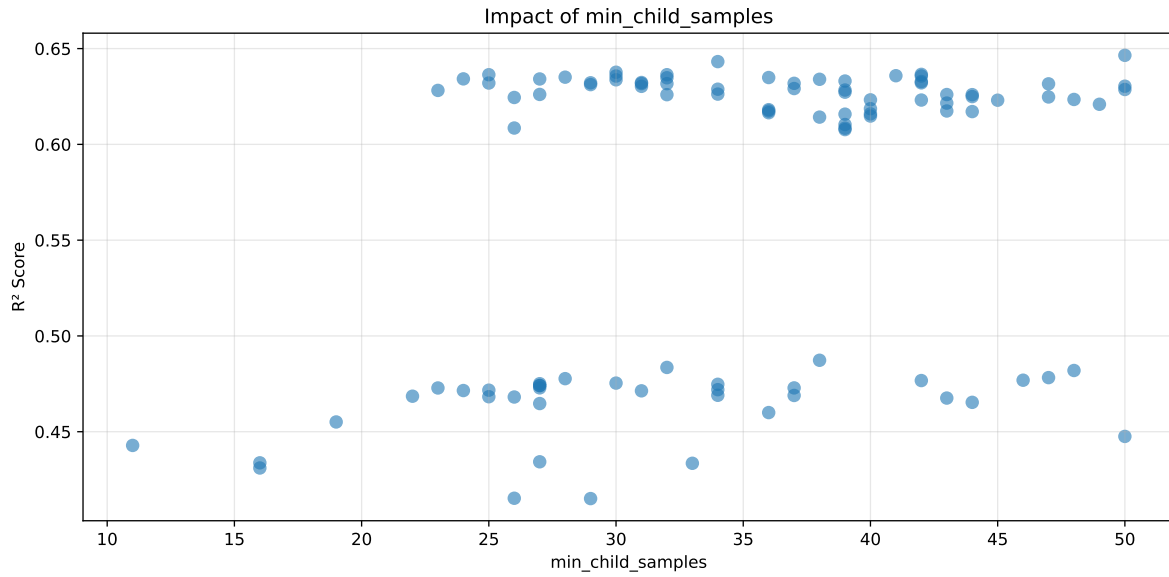
r_THI 2 r_NH3_st_8h_median 2 r_NH3_st_12h_rms 2 r_umidity_st_24h_std
 2 r_NH3_st_12h_min 2 r_temperature_st_12h_max 2 r_umidity_st_24h_rms 2
 r_temperature_st_8h_max 2 r_umidity_st_12h_median 1 r_NH3_st_8h_min 1 r_tem-
 perature_st_12h_rms 1 r_temperature 1 r_umidity_st_8h_std 1 r_umidity_st_8h_rms
 1 r_NH3_st_12h_diff_last 1 r_temperature_st_8h_median 1 r_NH3_st_12h_max 1
 r_NH3_st_12h_median 1 r_umidity_st_24h_median 0 r_umidity_st_24h_diff_last
 0 r_NH3_st_24h_diff_last 0 r_umidity_st_12h_rms 0 r_umidity_st_12h_mean 0
 r_NH3_st_8h_mean 0 r_temperature_st_24h_diff_last 0 r_temperature_st_24h_min 0
 r_temperature_st_24h_rms 0 r_temperature_st_24h_mean 0 r_temperature_st_12h_me-
 dian 0 r_temperature_st_12h_mean 0 r_NH3_st_8h_max 0 r_temperature_st_8h_rms 0
 r_temperature_st_8h_mean 0 r_NH3_st_8h_rms 0





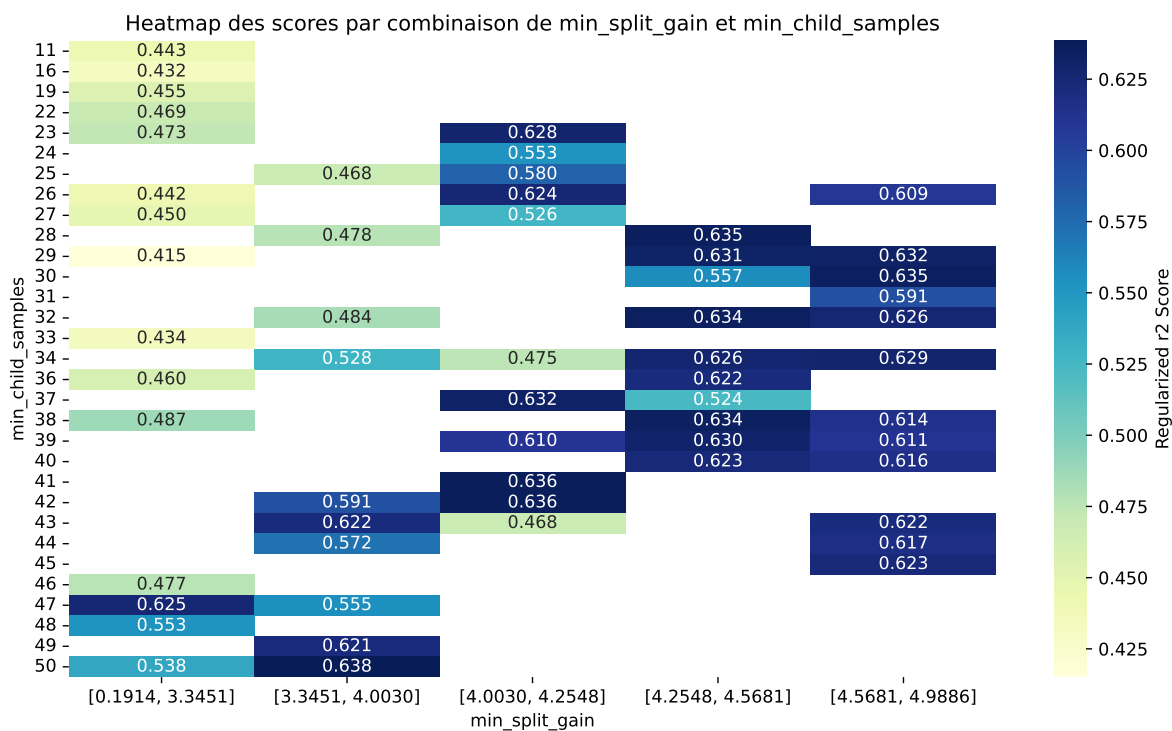
Paramètres avec >5% d'importance : ['min_split_gain', 'min_child_samples', 'reg_alpha']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6765	0.6465	0.6295	0.0124	0.6465	0.6498	0.0033	1.0052	1.0464
2	0.6747	0.6432	0.6221	0.0116	0.6432	0.6396	-0.0036	0.9944	1.0489
3	0.668	0.6377	0.6179	0.012	0.6377	0.6364	-0.0013	0.998	1.0476
4	0.6669	0.6365	0.6153	0.0108	0.6365	0.6343	-0.0022	0.9965	1.0477
5	0.6682	0.6364	0.6164	0.0118	0.6364	0.6376	0.0012	1.0019	1.0499
6	0.6662	0.6364	0.615	0.013	0.6364	0.6379	0.0015	1.0024	1.0468
7	0.6662	0.6358	0.6144	0.0082	0.6358	0.6355	-0.0004	0.9994	1.0477
8	0.6671	0.6357	0.6155	0.011	0.6357	0.6404	0.0047	1.0074	1.0495
9	0.6656	0.6357	0.6138	0.0113	0.6357	0.6354	-0.0003	0.9995	1.047
10	0.6665	0.6351	0.6154	0.0118	0.6351	0.6391	0.0041	1.0064	1.0495
---	---	---	---	---	---	---	---	---	---
100	0.76	0.7025	0.6854	0.0128	0.4151	0.7132	0.0107	1.0153	1.0818

Hyperparameters:

Rank 1: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.05072063670855887, 'reg__num_leaves': 63, 'reg__subsample': 0.8102484051496006, 'reg__colsample_bytree': 0.8984398362778085, 'reg__min_child_samples': 50, 'reg__reg_alpha': 4.899339658199369, 'reg__reg_lambda': 19.825946061617742, 'reg__min_split_gain': 3.380990081044956, 'reg__path_smooth': 3.697040697646729, 'reg__min_child_weight':

0.0012134831639281354, 'reg__feature_fraction': 0.8992962062861402, 'reg__bagging_fraction': 0.5990923596824277, 'reg__bagging_freq': 1, 'reg__max_bin': 65, 'reg__min_data_per_group': 98}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.06526972351641866, 'reg__num_leaves': 57, 'reg__subsample': 0.8884387429561682, 'reg__colsample_bytree': 0.796183102233726, 'reg__min_child_samples': 34, 'reg__reg_alpha': 1.266781858369729, 'reg__reg_lambda': 20.05496396500703, 'reg__min_split_gain': 3.6878650232825647, 'reg__path_smooth': 2.903243118324272, 'reg__min_child_weight': 0.001652143779170791, 'reg__feature_fraction': 0.6983002467495133, 'reg__bagging_fraction': 0.6139808521341316, 'reg__bagging_freq': 4, 'reg__max_bin': 134, 'reg__min_data_per_group': 118}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.06609319990250202, 'reg__num_leaves': 55, 'reg__subsample': 0.896347052526297, 'reg__colsample_bytree': 0.7702772575692196, 'reg__min_child_samples': 30, 'reg__reg_alpha': 0.7050452740534148, 'reg__reg_lambda': 29.362083847038143, 'reg__min_split_gain': 4.354124311583572, 'reg__path_smooth': 1.6046248915804686, 'reg__min_child_weight': 0.0020250529633540747, 'reg__feature_fraction': 0.7150437783626868, 'reg__bagging_fraction': 0.6719422930928262, 'reg__bagging_freq': 3, 'reg__max_bin': 120, 'reg__min_data_per_group': 82}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.058290436871498194, 'reg__num_leaves': 56, 'reg__subsample': 0.5826339244020025, 'reg__colsample_bytree': 0.8302812084081063, 'reg__min_child_samples': 42, 'reg__reg_alpha': 0.23680692245050072, 'reg__reg_lambda': 22.20094208089216, 'reg__min_split_gain': 4.074478775203488, 'reg__path_smooth': 1.9356890507680617, 'reg__min_child_weight': 0.00291891582178731, 'reg__feature_fraction': 0.8463931504325649, 'reg__bagging_fraction': 0.6134633501318559, 'reg__bagging_freq': 4, 'reg__max_bin': 112, 'reg__min_data_per_group': 75}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.07214588831062337, 'reg__num_leaves': 41, 'reg__subsample': 0.63404235300264, 'reg__colsample_bytree': 0.7547173569621296, 'reg__min_child_samples': 32, 'reg__reg_alpha': 0.8213737195517956, 'reg__reg_lambda': 24.002535041554978, 'reg__min_split_gain': 4.454064661992167, 'reg__path_smooth': 1.5360958128899997, 'reg__min_child_weight': 0.014727757096584821, 'reg__feature_fraction': 0.8211702481626526, 'reg__bagging_fraction': 0.6552216375958251, 'reg__bagging_freq': 3, 'reg__max_bin': 193, 'reg__min_data_per_group': 53}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.04625892372561451, 'reg__num_leaves': 59, 'reg__subsample': 0.5482813137936083, 'reg__colsample_bytree': 0.8830183852395097, 'reg__min_child_samples': 25, 'reg__reg_alpha': 3.365444988395282, 'reg__reg_lambda': 29.834549568428585, 'reg__min_split_gain': 4.139635879368724, 'reg__path_smooth': 1.2965758513010257, 'reg__min_child_weight':

0.007066034681062749, 'reg__feature_fraction': 0.8313138667203678, 'reg__bagging_fraction': 0.7121261950176572, 'reg__bagging_freq': 2, 'reg__max_bin': 163, 'reg__min_data_per_group': 104}

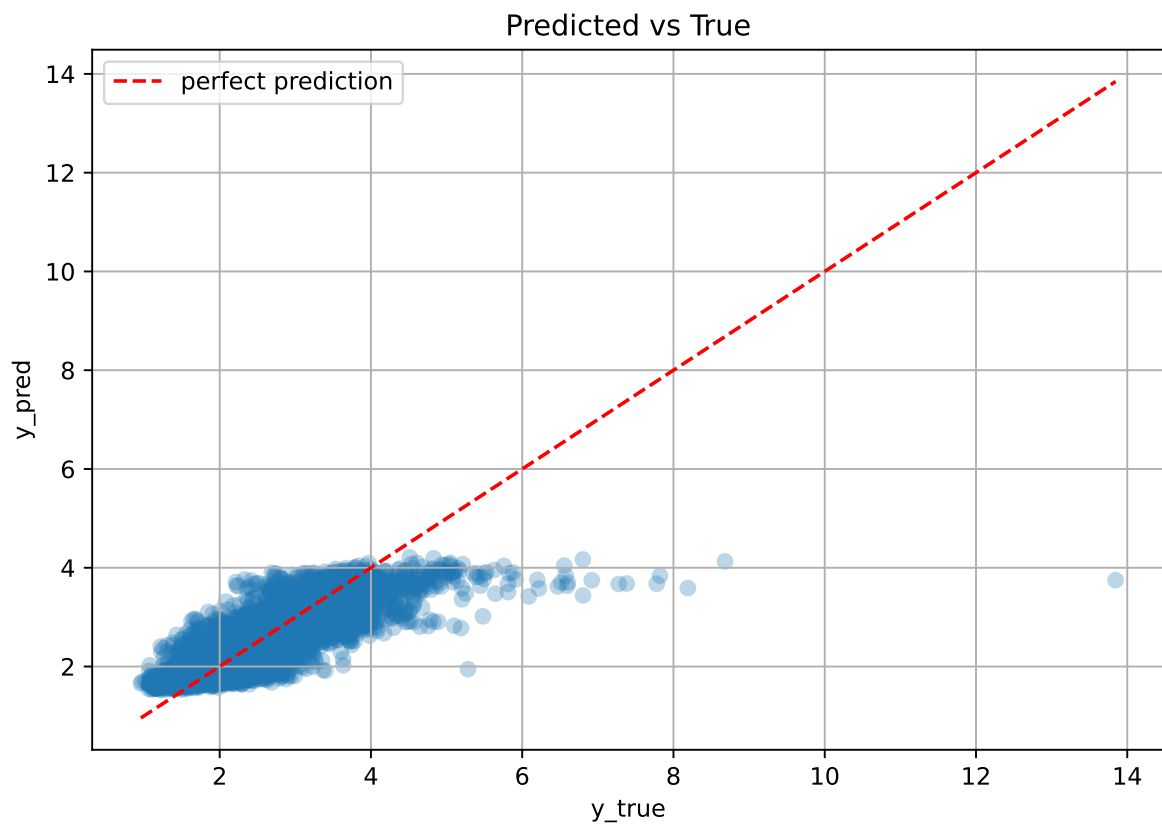
Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.05761466901979272, 'reg__num_leaves': 47, 'reg__subsample': 0.8045023117306389, 'reg__colsample_bytree': 0.8226177809639637, 'reg__min_child_samples': 41, 'reg__reg_alpha': 0.2709858516769028, 'reg__reg_lambda': 20.897859847756003, 'reg__min_split_gain': 4.040034089461892, 'reg__path_smooth': 2.228908963930747, 'reg__min_child_weight': 0.0010760798160506016, 'reg__feature_fraction': 0.8524248982730935, 'reg__bagging_fraction': 0.6104485769550927, 'reg__bagging_freq': 5, 'reg__max_bin': 106, 'reg__min_data_per_group': 109}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.09231345080058155, 'reg__num_leaves': 58, 'reg__subsample': 0.5859971823794641, 'reg__colsample_bytree': 0.8030338663368541, 'reg__min_child_samples': 42, 'reg__reg_alpha': 0.14810481486211202, 'reg__reg_lambda': 21.700059476082593, 'reg__min_split_gain': 4.013427969283251, 'reg__path_smooth': 1.8571052399999335, 'reg__min_child_weight': 0.0010058458955332556, 'reg__feature_fraction': 0.8417252569775667, 'reg__bagging_fraction': 0.6023736200802986, 'reg__bagging_freq': 4, 'reg__max_bin': 122, 'reg__min_data_per_group': 76}

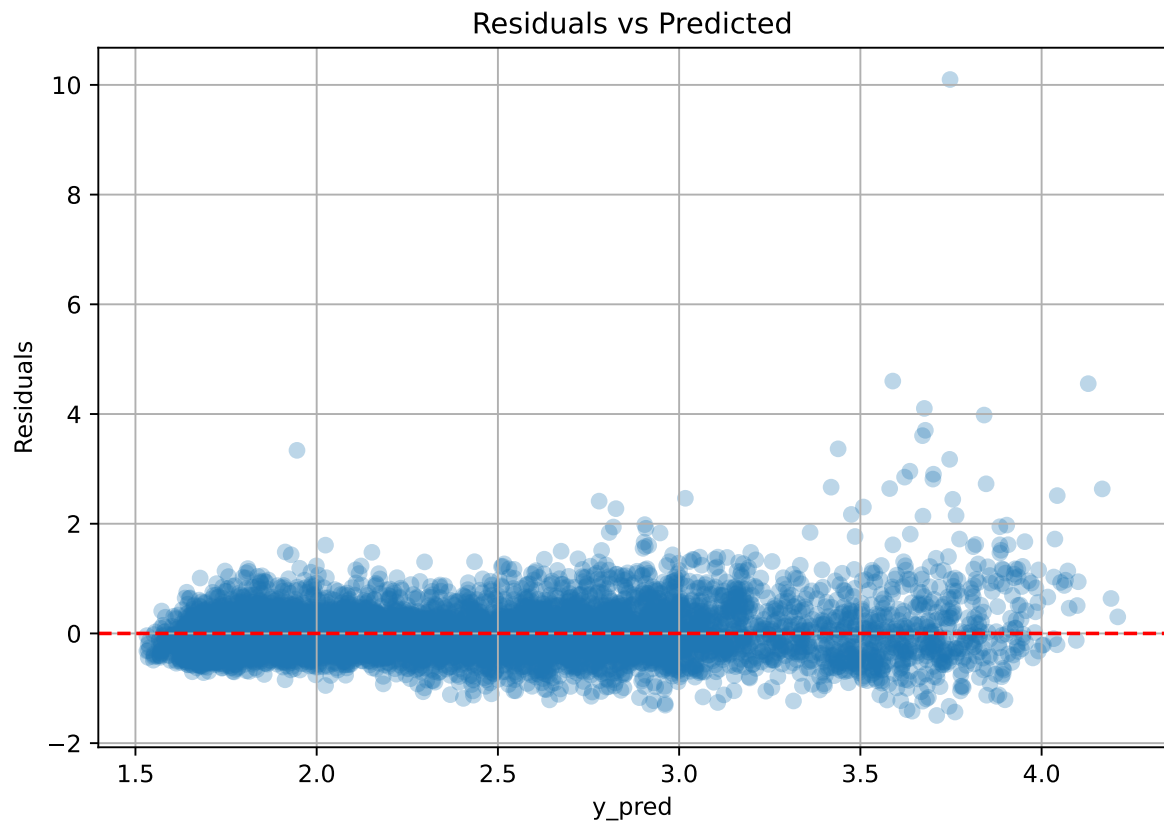
Rank 9: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.051216433769108734, 'reg__num_leaves': 36, 'reg__subsample': 0.8985651207035914, 'reg__colsample_bytree': 0.788105722569862, 'reg__min_child_samples': 30, 'reg__reg_alpha': 0.658969861806761, 'reg__reg_lambda': 24.146169231068125, 'reg__min_split_gain': 4.586249197631495, 'reg__path_smooth': 2.5859889446173794, 'reg__min_child_weight': 0.0066570933319246755, 'reg__feature_fraction': 0.7116506973303063, 'reg__bagging_fraction': 0.6523794150115074, 'reg__bagging_freq': 2, 'reg__max_bin': 149, 'reg__min_data_per_group': 94}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.06470788503040878, 'reg__num_leaves': 55, 'reg__subsample': 0.896957313857714, 'reg__colsample_bytree': 0.7680786752335268, 'reg__min_child_samples': 28, 'reg__reg_alpha': 1.353243377936438, 'reg__reg_lambda': 29.68330911140496, 'reg__min_split_gain': 4.427542508557134, 'reg__path_smooth': 1.5660863447461013, 'reg__min_child_weight': 0.0021294061351379407, 'reg__feature_fraction': 0.7146029402472257, 'reg__bagging_fraction': 0.6764106536525911, 'reg__bagging_freq': 3, 'reg__max_bin': 174, 'reg__min_data_per_group': 83}

2.4.2 Scatter



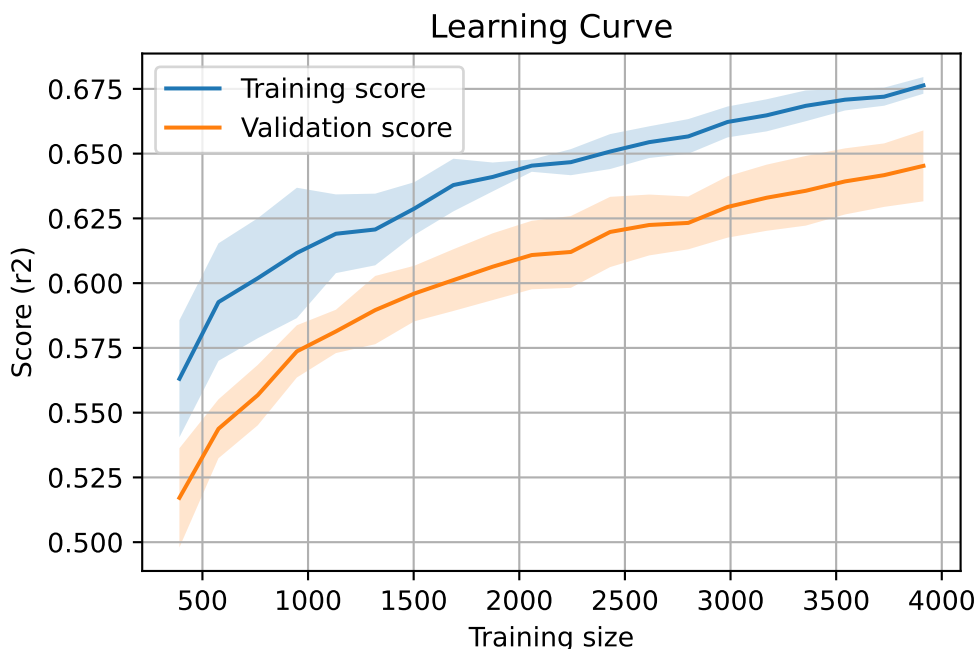
2.4.3 Scatter residuals



2.4.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396



2.5 Model : CatBoost

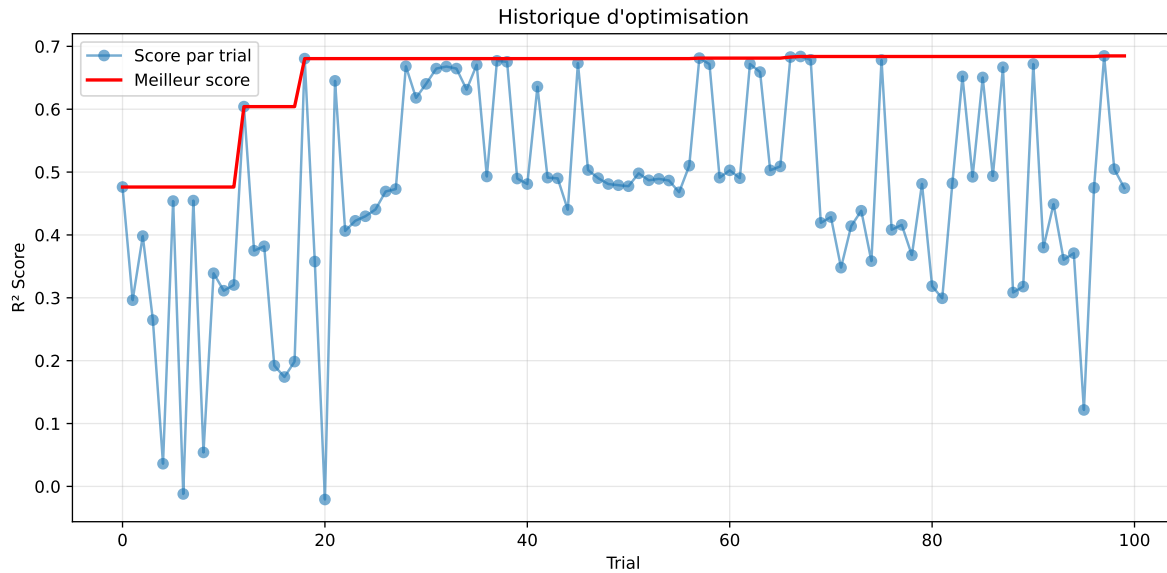
2.5.1 Gridsearch

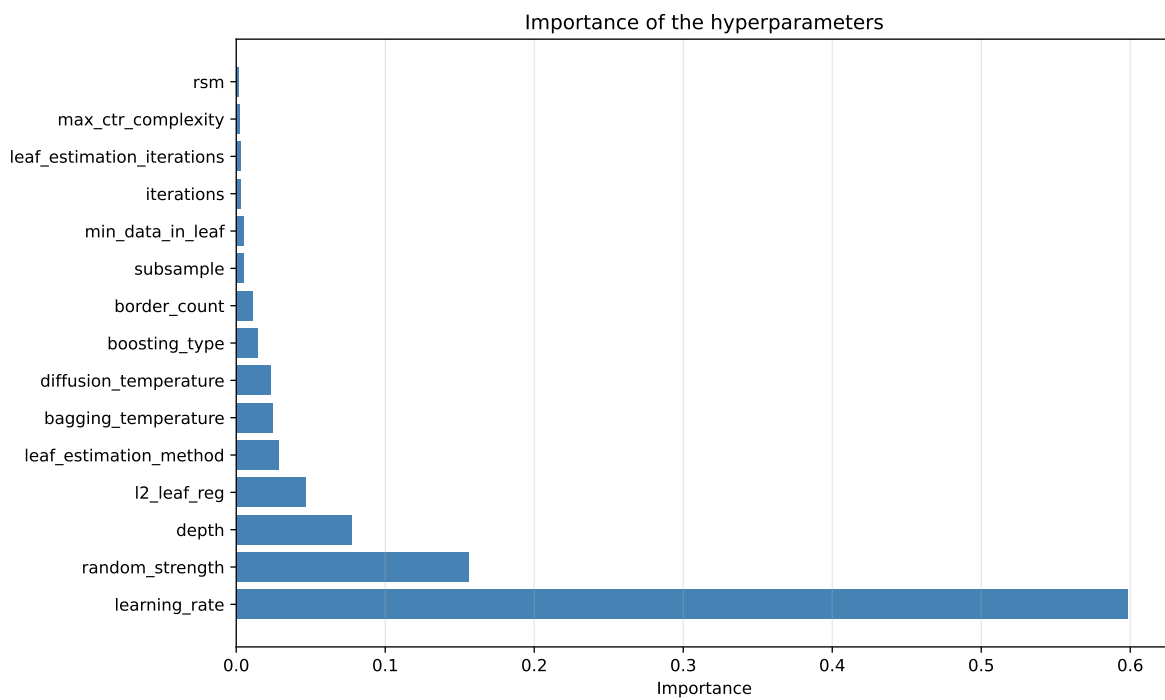
Distribution y_train vs y_test: y_train: mean=2.350, std=0.793, min=0.959, max=8.682
y_test: mean=2.359, std=0.815, min=0.973, max=13.845

===== TOP Importance FEATURES =====

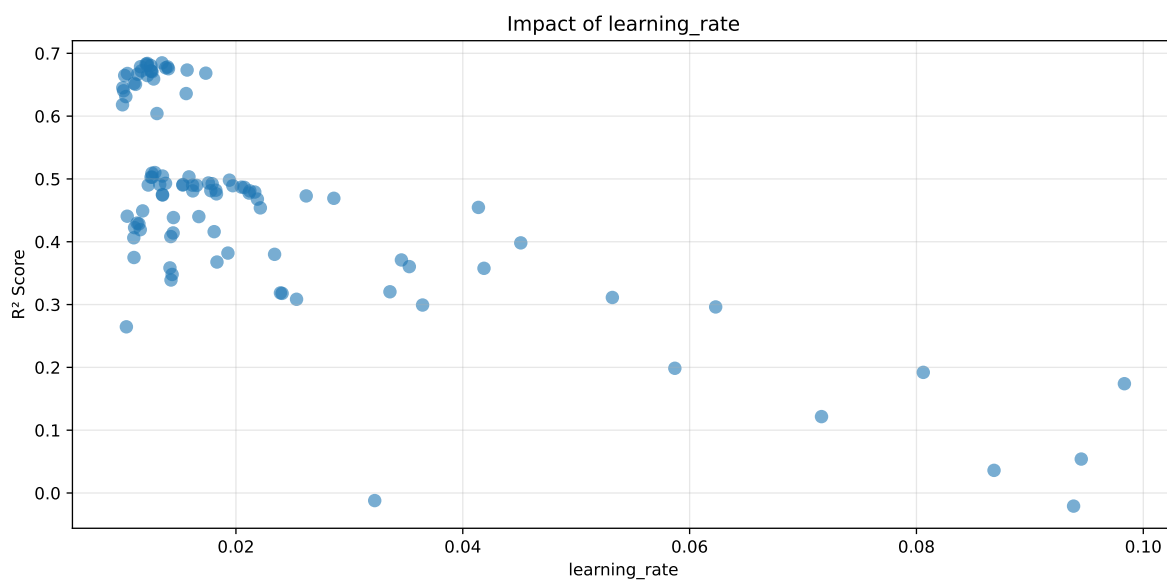
feature importance r_temperature_st_12h_slope 9.149422 r_umidity 8.589621 r_temperature_st_24h_slope 7.947601 r_CO2 7.327613 r_temperature_st_8h_slope 5.739938 r_umidity_st_12h_slope 4.067712 r_temperature_st_12h_std 3.743712 r_umidity_st_8h_slope 3.323957 r_temperature_st_24h_diff_last 2.225452 r_NH3_st_12h_std 1.798702 r_NH3_st_24h_min 1.767079 r_umidity_st_12h_max 1.726514 r_temperature_st_8h_diff_last 1.616995 r_umidity_st_12h_min 1.462387 r_NH3_st_8h_std 1.385633 r_temperature_st_8h_std 1.339600 r_NH3_st_24h_std 1.305044 r_umidity_st_8h_min 1.302680 r_umidity_st_8h_mean 1.266515 r_temperature_st_12h_diff_last 1.261669 r_umidity_st_8h_median 1.191195 r_umidity_st_8h_rms 1.171092 r_NH3_st_24h_max 1.023422 r_umidity_st_8h_max 1.021320 r_umidity_st_12h_rms 0.992590 r_NH3 0.966269 r_temperature_st_24h_max 0.921891 r_NH3_st_24h_median 0.909451 r_NH3_st_12h_max 0.811876 r_temperature_st_24h_mean 0.789260 r_umidity_st_24h_slope 0.777765 r_temperature_st_24h_rms 0.773026 r_NH3_st_24h_slope 0.732118 r_umidity_st_24h_rms 0.730214 r_NH3_st_12h_min 0.712846 r_umidity_st_12h_mean 0.709077

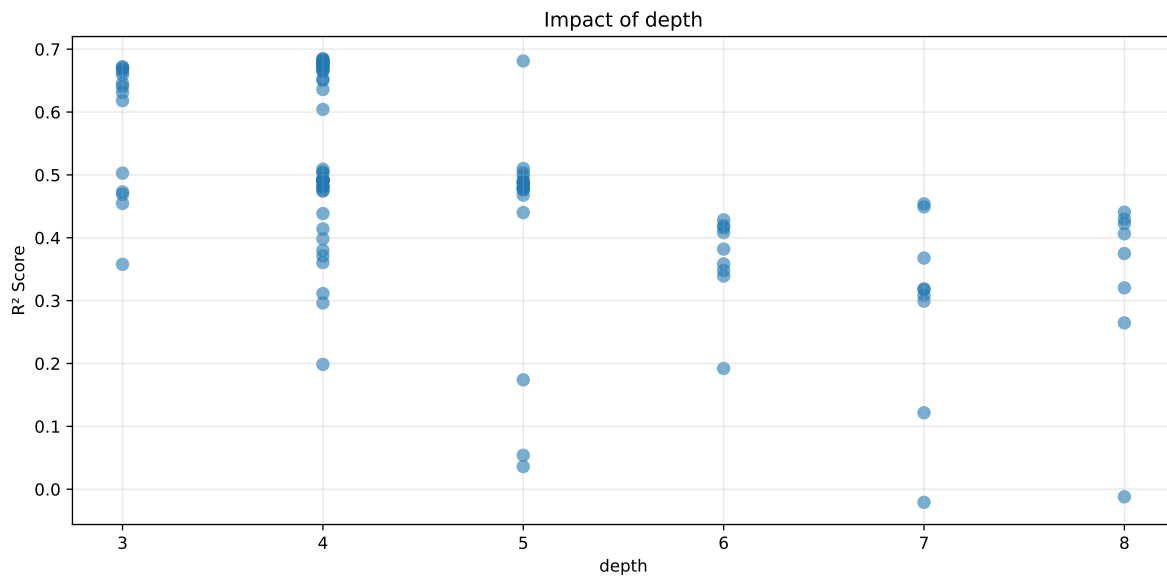
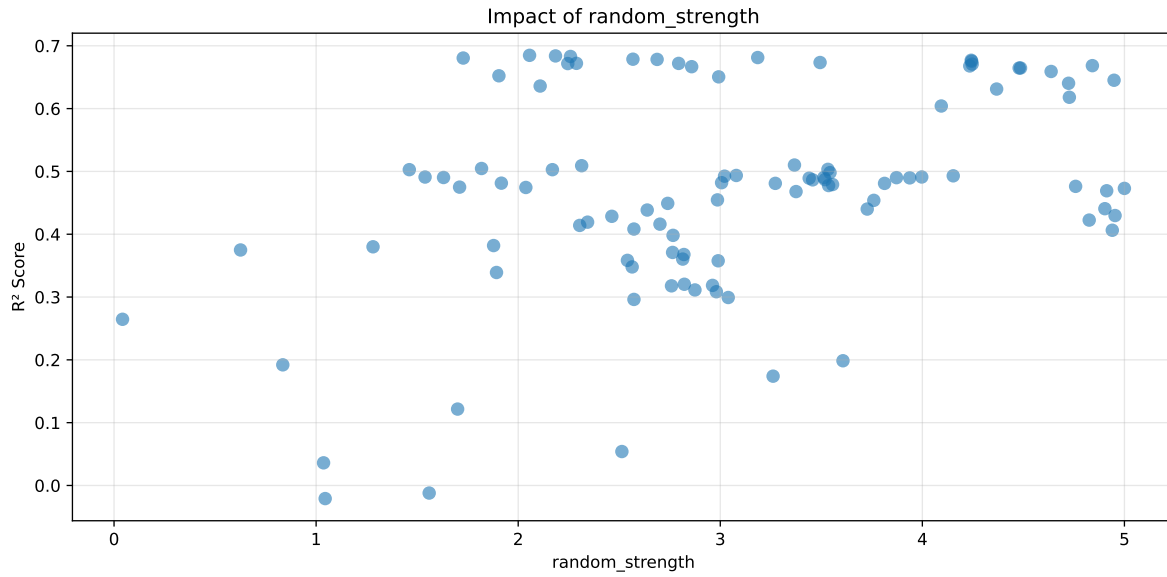
r_NH3_st_12h_slope 0.677440 r_umidity_st_24h_max 0.668426 r_umidity_st_24h_mean
 0.658237 r_NH3_st_24h_mean 0.651906 r_NH3_st_8h_max 0.634570 r_umid-
 ity_st_24h_min 0.614248 r_temperature_st_24h_median 0.604236 r_NH3_st_24h_rms
 0.591673 r_NH3_st_12h_rms 0.579293 r_NH3_st_12h_median 0.543235 r_NH3_st_8h_me-
 dian 0.531955 r_NH3_st_12h_mean 0.529531 r_NH3_st_8h_slope 0.519711 r_umid-
 ity_st_12h_std 0.514588 r_temperature_st_12h_min 0.499562 r_umidity_st_24h_me-
 dian 0.476813 r_temperature 0.475746 r_temperature_st_12h_max 0.454113 r_tem-
 perature_st_8h_min 0.453698 r_NH3_st_8h_min 0.441078 r_NH3_st_8h_mean
 0.433637 r_NH3_st_8h_rms 0.398292 r_temperature_st_12h_mean 0.398050 r_tempera-
 ture_st_8h_max 0.387470 r_umidity_st_8h_diff_last 0.379028 r_umidity_st_24h_diff_last
 0.375504 r_temperature_st_24h_std 0.350342 r_THI 0.319814 r_temperature_st_8h_me-
 dian 0.318319 r_NH3_st_12h_diff_last 0.317660 r_umidity_st_12h_diff_last 0.316444
 r_temperature_st_12h_median 0.303479 r_temperature_st_12h_rms 0.299302 r_NH3_st_24h_diff_last
 0.280356 r_umidity_st_24h_std 0.269595 r_umidity_st_8h_std 0.252922 r_tempera-
 ture_st_8h_mean 0.246538 r_umidity_st_12h_median 0.232047 r_temperature_st_8h_rms
 0.193550 r_NH3_st_8h_diff_last 0.123678 r_temperature_st_24h_min 0.102655





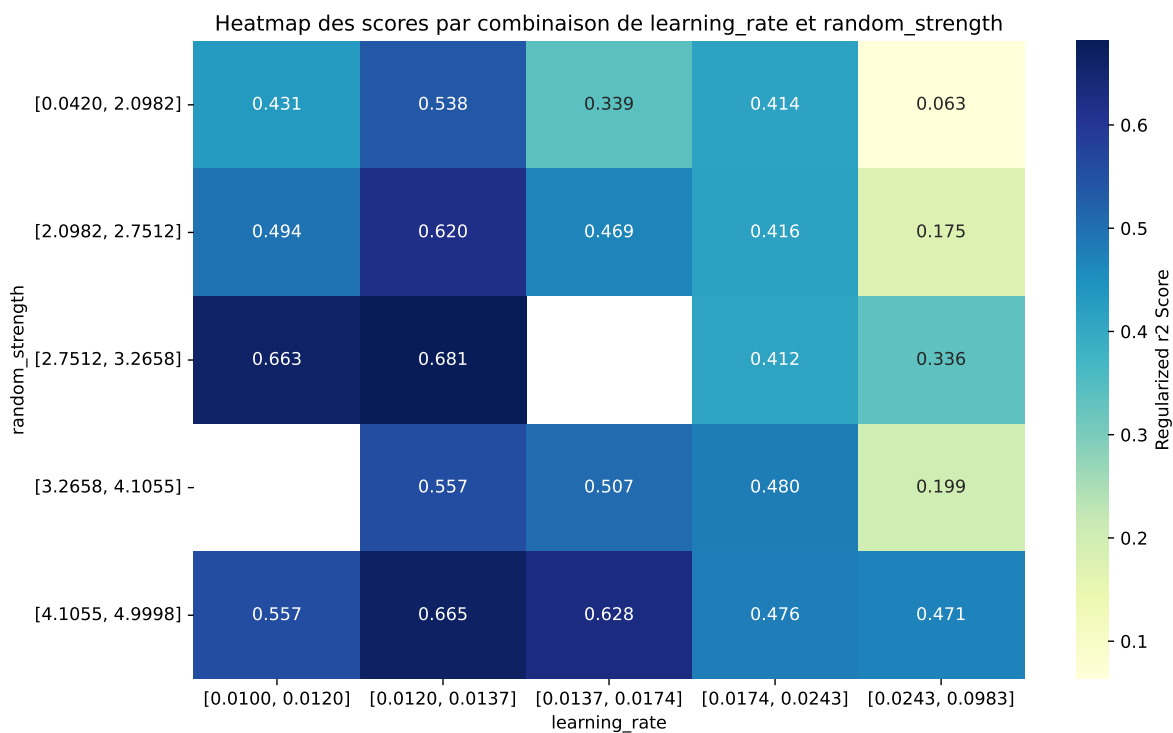
Paramètres avec >5% d'importance : ['learning_rate', 'random_strength', 'depth']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7189	0.6848	0.6671	0.0112	0.6848	0.6777	-0.0071	0.9897	1.0497
2	0.7179	0.6839	0.6666	0.0115	0.6839	0.6772	-0.0067	0.9903	1.0497
3	0.7169	0.683	0.6661	0.0109	0.683	0.6757	-0.0073	0.9893	1.0496
4	0.7153	0.6813	0.6635	0.011	0.6813	0.673	-0.0083	0.9879	1.0499
5	0.7144	0.6805	0.6631	0.0103	0.6805	0.6743	-0.0062	0.9909	1.0498
6	0.712	0.6786	0.6623	0.0103	0.6786	0.6708	-0.0078	0.9885	1.0491
7	0.7115	0.6783	0.6614	0.0103	0.6783	0.6684	-0.0099	0.9854	1.0489
8	0.7086	0.677	0.6591	0.0102	0.677	0.6687	-0.0083	0.9878	1.0467
9	0.7079	0.6754	0.6575	0.009	0.6754	0.6686	-0.0068	0.9899	1.0481
10	0.7056	0.6734	0.6573	0.0109	0.6734	0.6643	-0.009	0.9866	1.0478
---	---	---	---	---	---	---	---	---	---
100	0.9744	0.8085	0.7847	0.0068	-0.0209	0.8073	-0.0013	0.9984	1.2052

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.013479417823562467, 'reg__l2_leaf_reg': 45.36711875779065, 'reg__bagging_temperature': 3.985988696009925, 'reg__random_strength': 2.0562436961688073, 'reg__subsample': 0.6171689184899579, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5248457251701627, 'reg__border_count': 84, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 1190.8400220403787}

Rank 2: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.0121904118497364, 'reg__l2_leaf_reg': 42.900566893670934, 'reg__bagging_temperature': 2.717852718472897, 'reg__random_strength': 2.184678787875363, 'reg__subsample': 0.6736938399328011, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5886178899182246, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 858.1794465384214}

Rank 3: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.012082237575081872, 'reg__l2_leaf_reg': 42.76894799680394, 'reg__bagging_temperature': 3.8377855358967987, 'reg__random_strength': 2.2587977981878247, 'reg__subsample': 0.6372130355702474, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5536312618345895, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 996.6268811337497}

Rank 4: {'reg__iterations': 900, 'reg__depth': 5, 'reg__learning_rate': 0.012171538455253388, 'reg__l2_leaf_reg': 32.29480441769281, 'reg__bagging_temperature': 4.482319676537198, 'reg__random_strength': 3.185318030791599, 'reg__subsample': 0.6320784659184859, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5538988109705244, 'reg__border_count': 90, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 827.7810054627512}

Rank 5: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.012498680877518735, 'reg__l2_leaf_reg': 41.012902078892814, 'reg__bagging_temperature': 3.753059803777457, 'reg__random_strength': 1.7279370738631923, 'reg__subsample': 0.6663900518052266, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5023055981929532, 'reg__border_count': 70, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1760.6489721330188}

Rank 6: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.011587068350389924, 'reg__l2_leaf_reg': 32.08687847944106, 'reg__bagging_temperature': 2.665359589906628, 'reg__random_strength': 2.5678069574009763, 'reg__subsample': 0.665140715638623, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.591522486970868, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 675.6593211239596}

Rank 7: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.013977239143888362, 'reg__l2_leaf_reg': 30.801333456687825, 'reg__bagging_temperature': 4.780463500315377, 'reg__random_strength': 2.6874255075421587, 'reg__subsample': 0.6479705039523731,

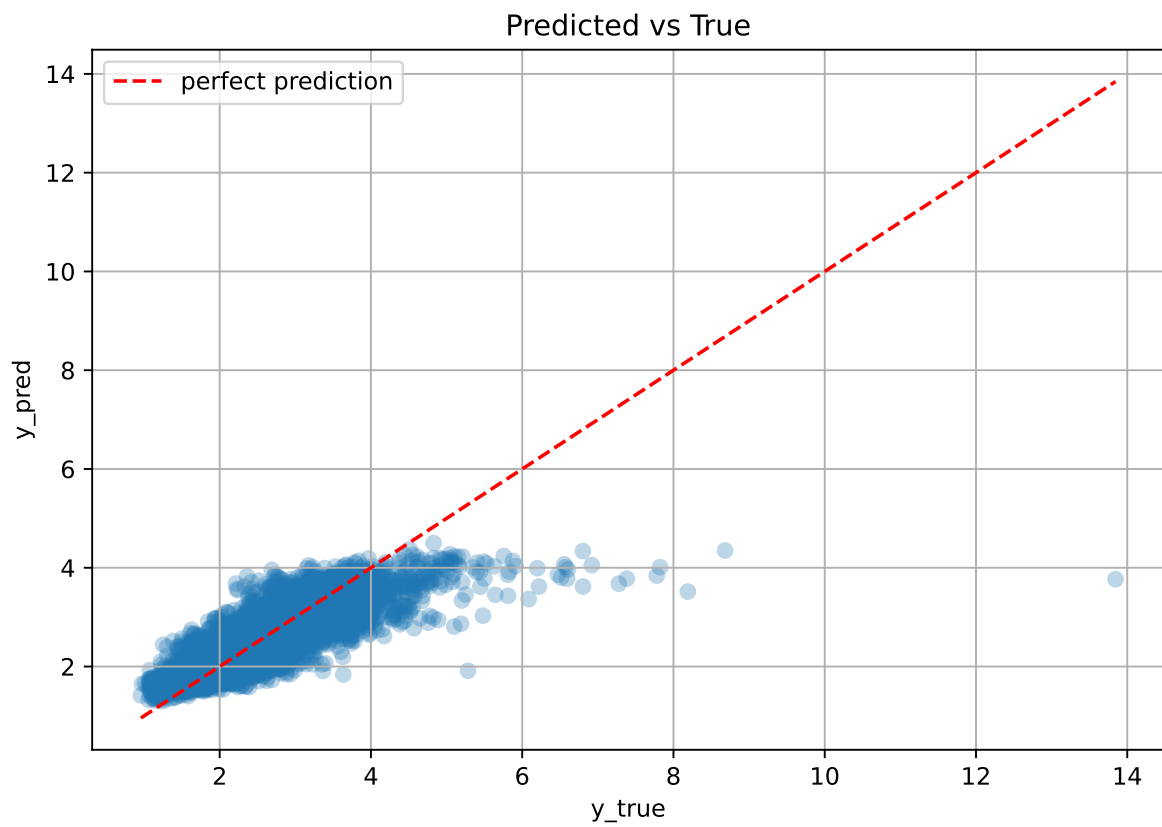
'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7800995601468613, 'reg__border_count': 84, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 1285.955405539974}

Rank 8: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.013807082393577745, 'reg__l2_leaf_reg': 35.80427886743057, 'reg__bagging_temperature': 4.029292114585035, 'reg__random_strength': 4.241769228808394, 'reg__subsample': 0.6727451526030044, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5333374157201342, 'reg__border_count': 102, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 1413.2230688489863}

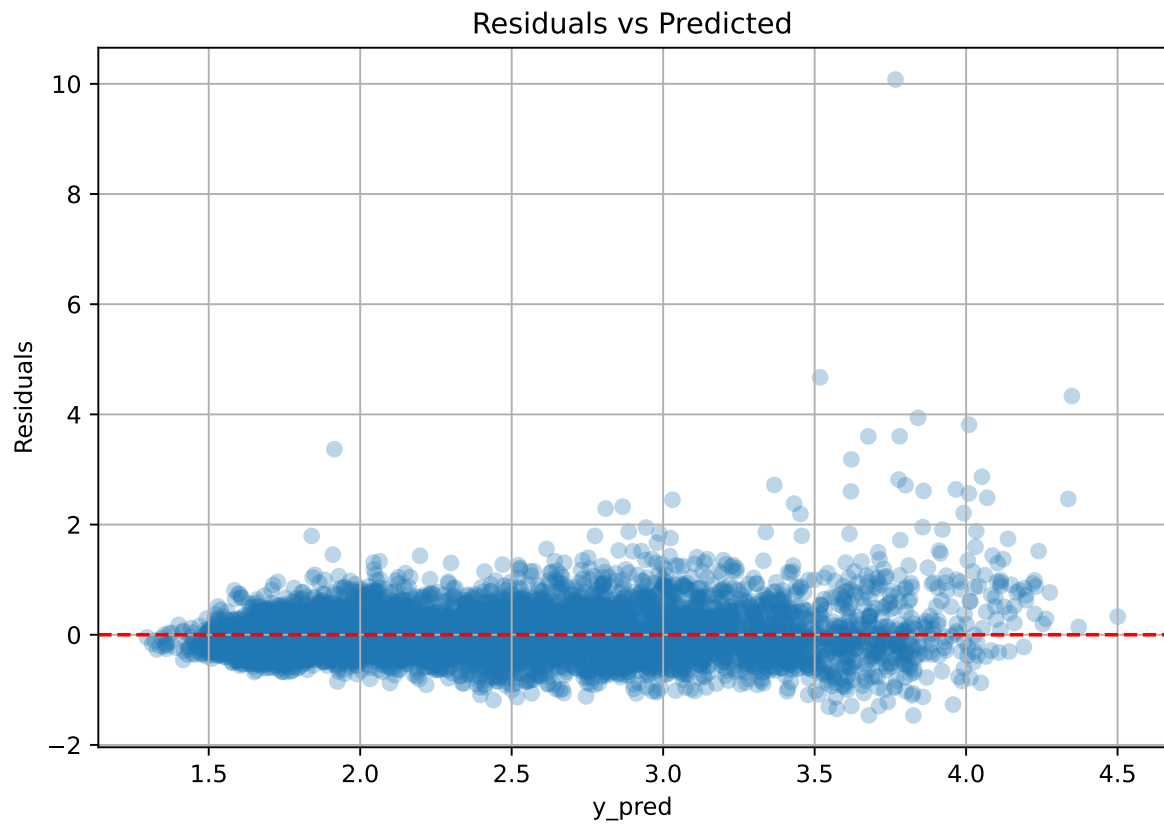
Rank 9: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.014050199950037735, 'reg__l2_leaf_reg': 36.21851988576677, 'reg__bagging_temperature': 4.106560651032003, 'reg__random_strength': 4.244177025778482, 'reg__subsample': 0.89328132777291, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5360503106043869, 'reg__border_count': 103, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 1538.335562281552}

Rank 10: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.0156957492615038, 'reg__l2_leaf_reg': 29.72184888772797, 'reg__bagging_temperature': 4.023194377763815, 'reg__random_strength': 3.4938755955020318, 'reg__subsample': 0.6611751389742409, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5321690021354523, 'reg__border_count': 111, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 83.45333107018519}

2.5.2 Scatter



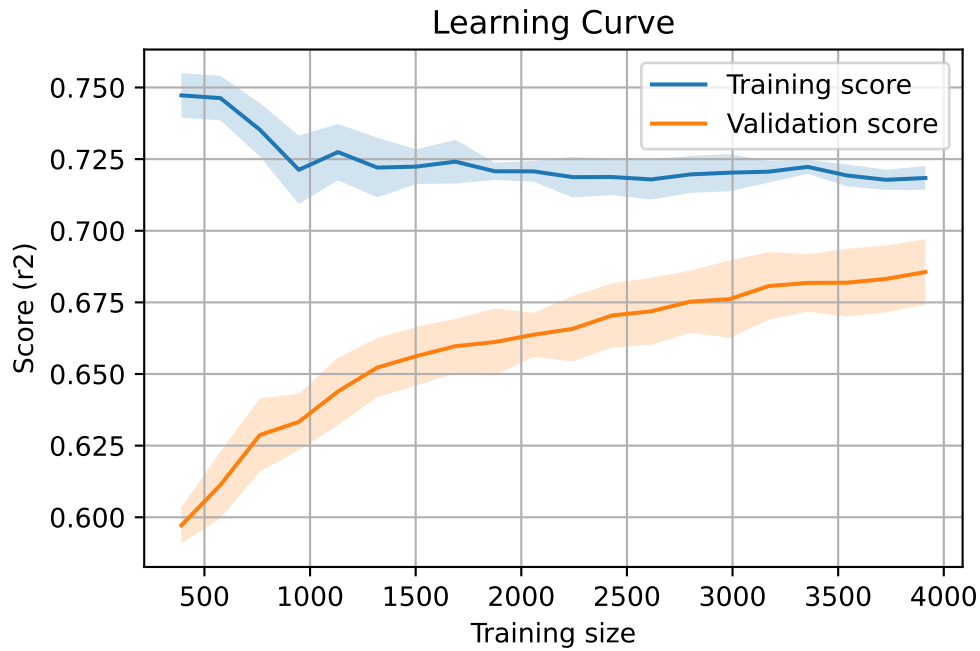
2.5.3 Scatter residuals



2.5.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396



2.6 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.584	0.564	0.009	0.541	0.545	-0.019	0.966	1.035	nan
LGBM	0.676	0.646	0.012	0.63	0.65	0.003	1.005	1.046	0.6465
CatBoost	0.719	0.685	0.011	0.667	0.678	-0.007	0.99	1.05	0.6848

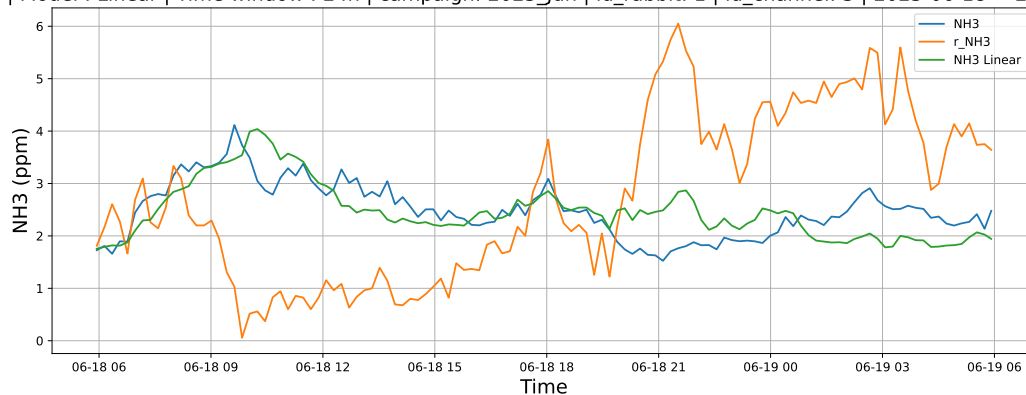
3 Plot

3.1 Standalone plots

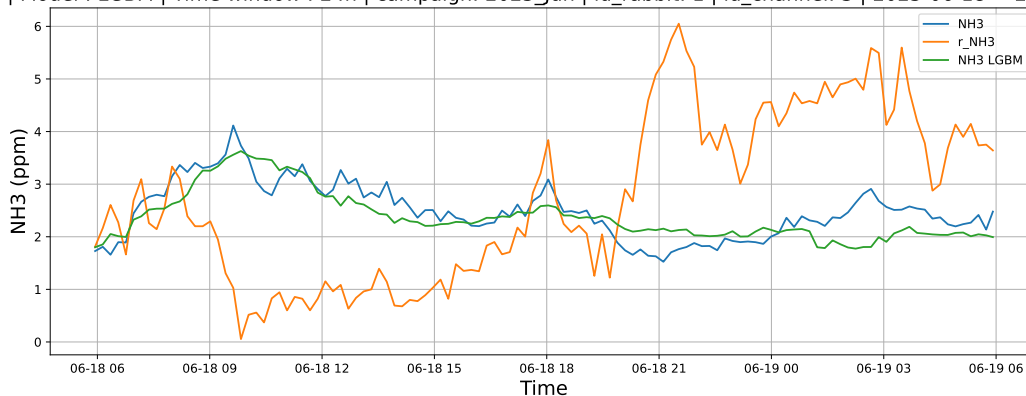
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

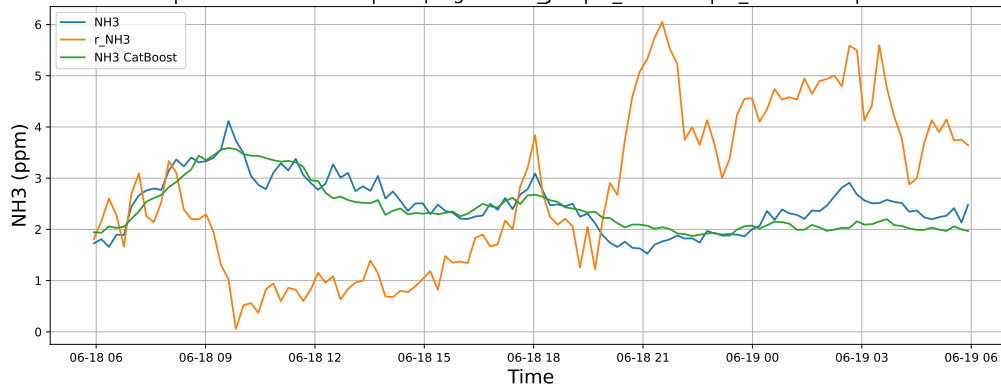
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

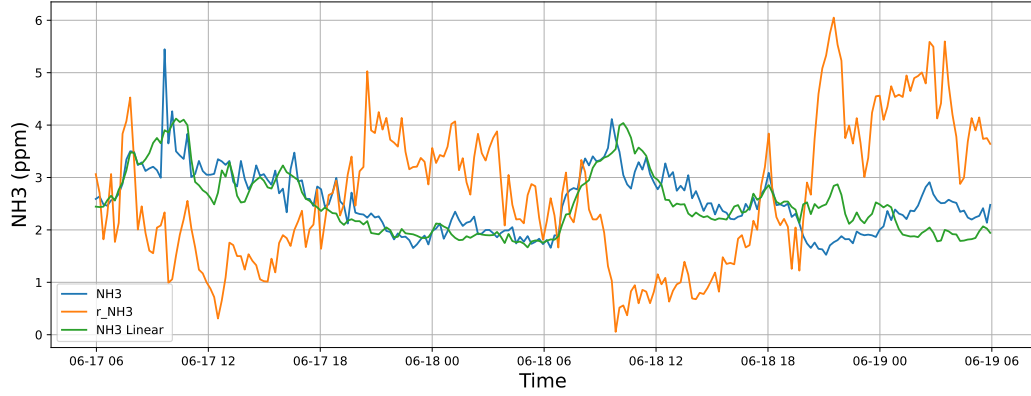


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

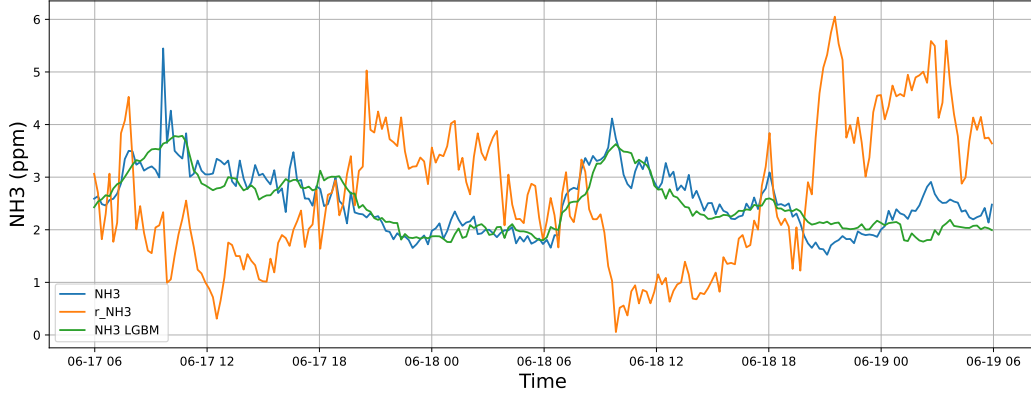


3.1.1.2 Time window : 48

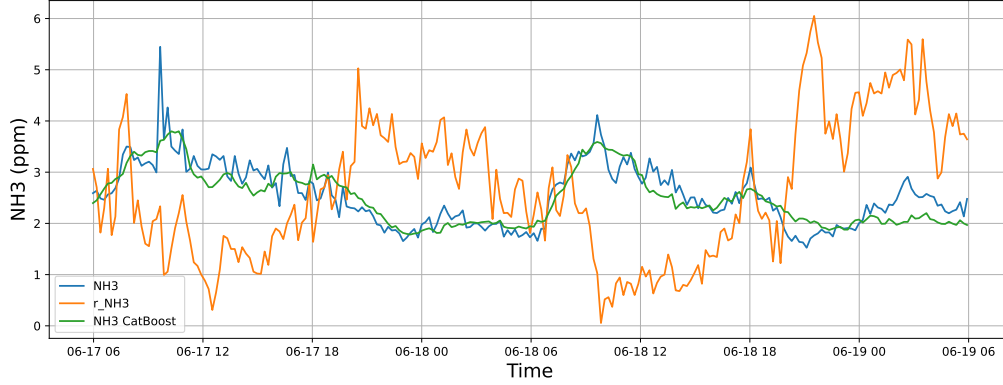
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

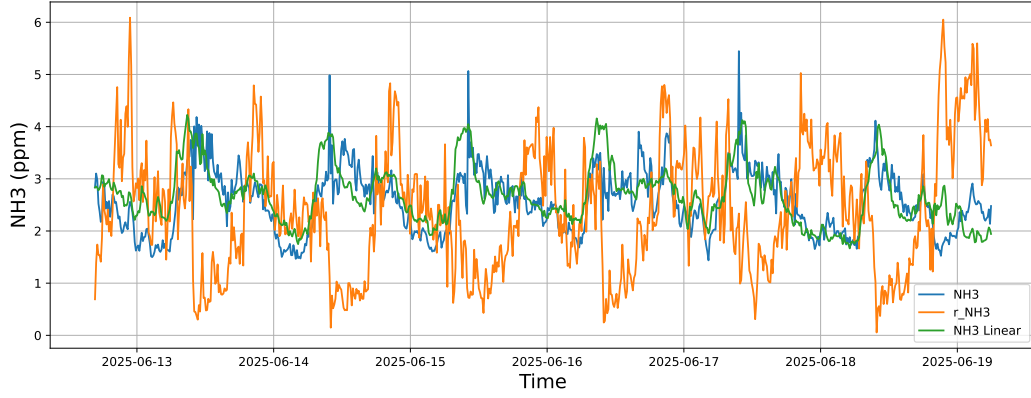


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

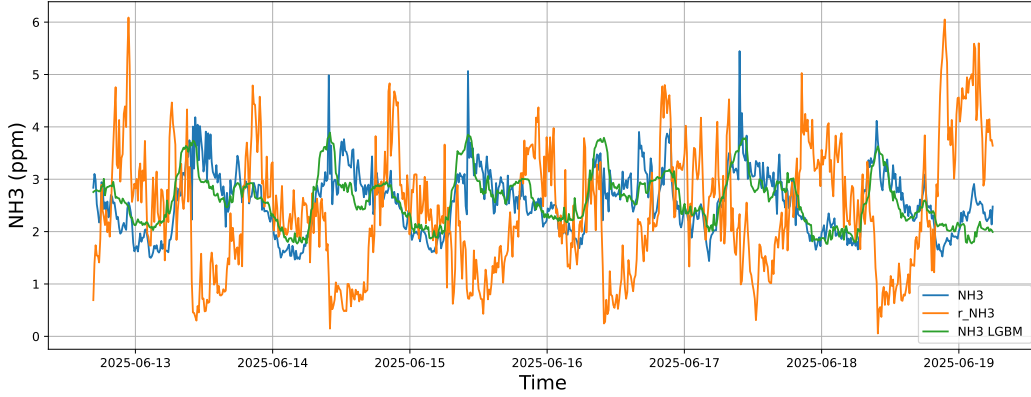


3.1.1.3 Time window : ALL

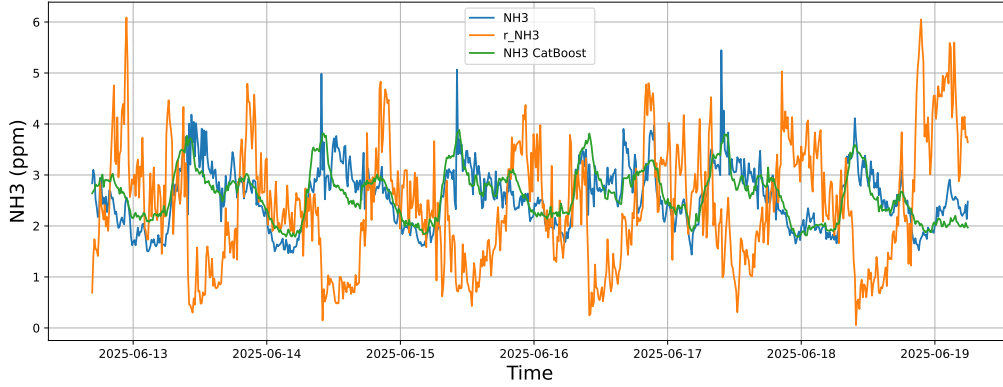
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



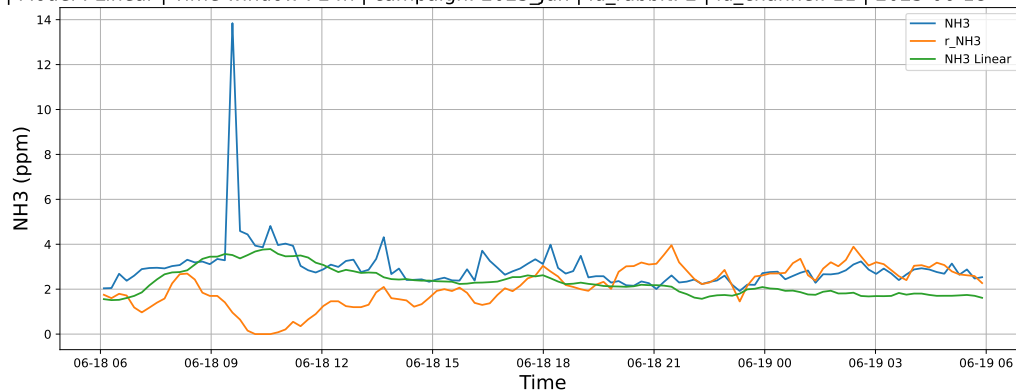
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



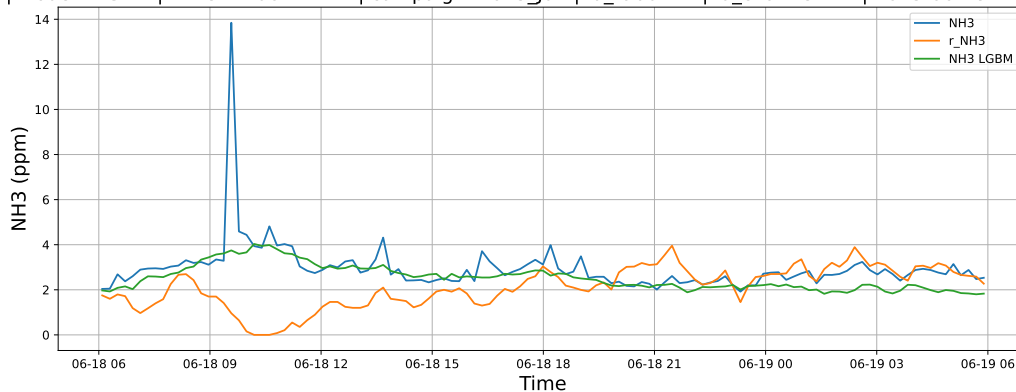
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

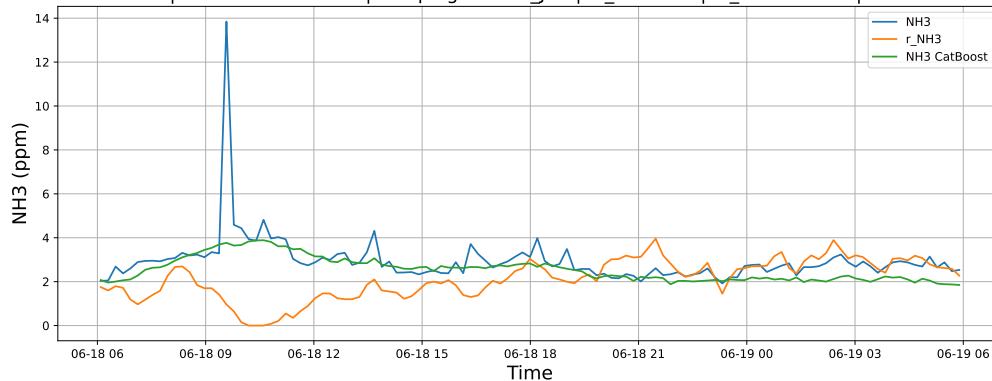
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

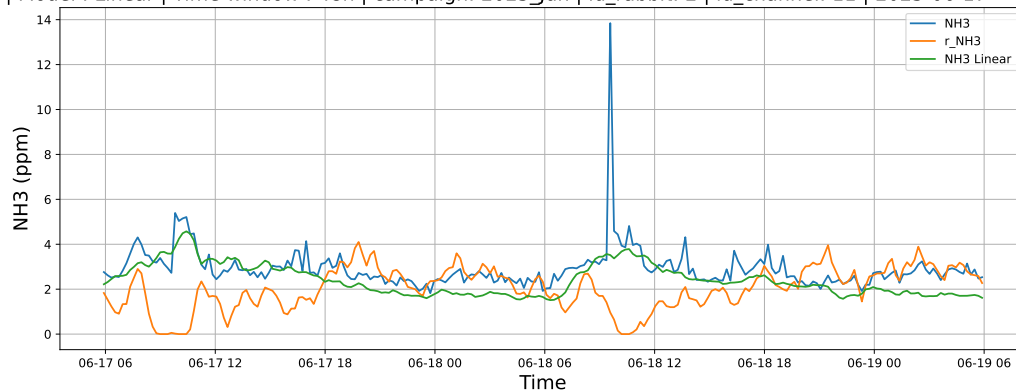


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

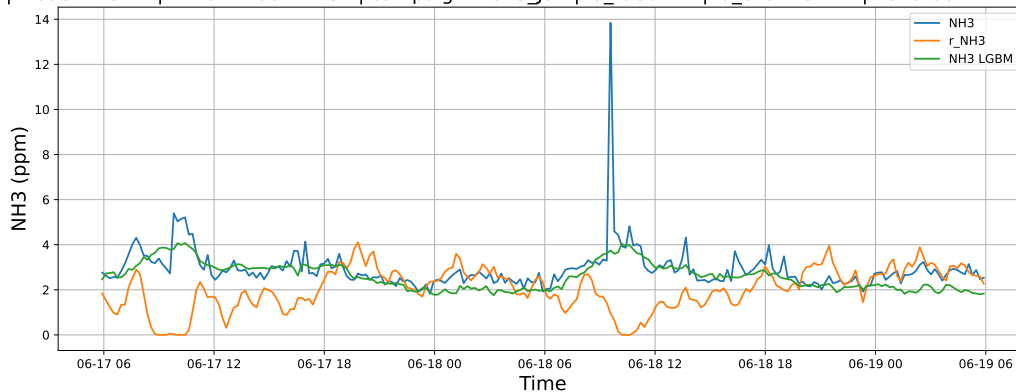


3.1.2.2 Time window : 48

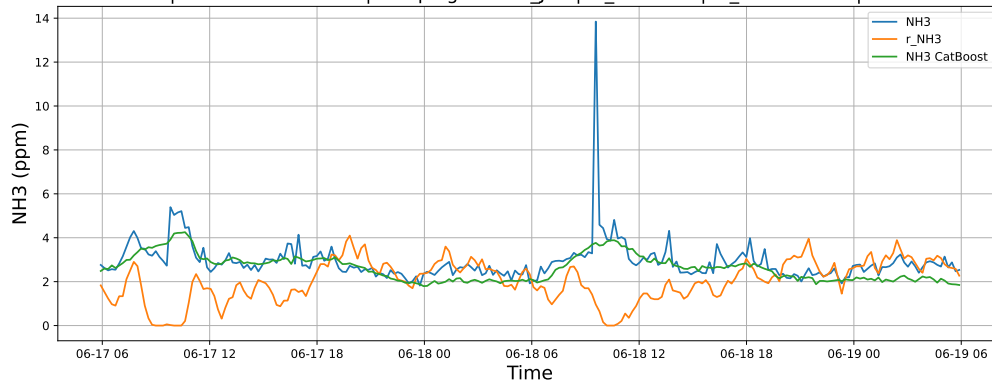
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

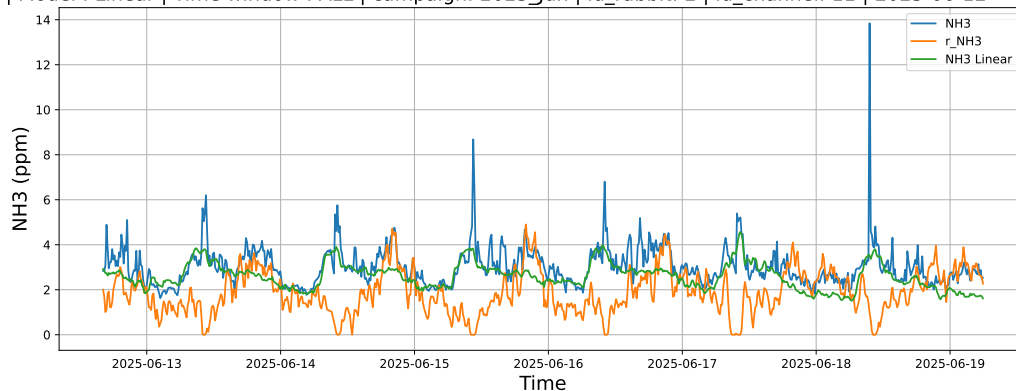


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

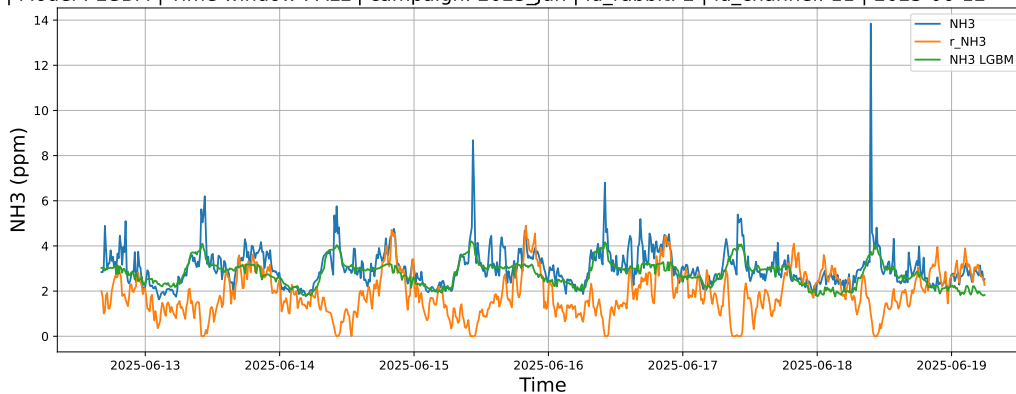


3.1.2.3 Time window : ALL

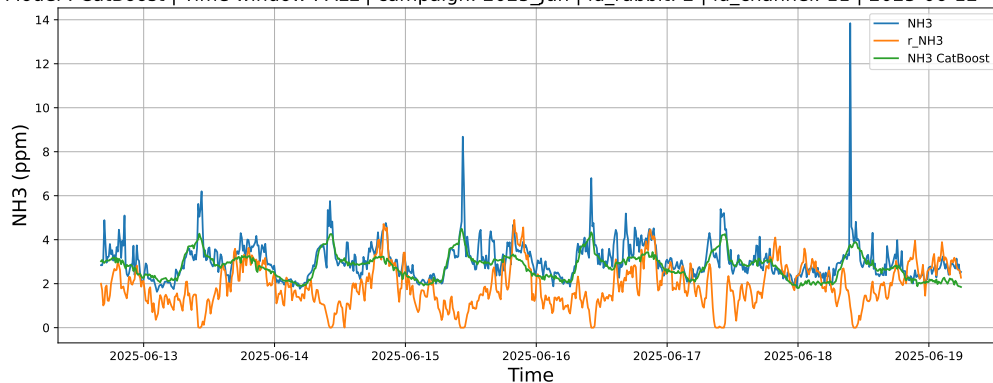
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



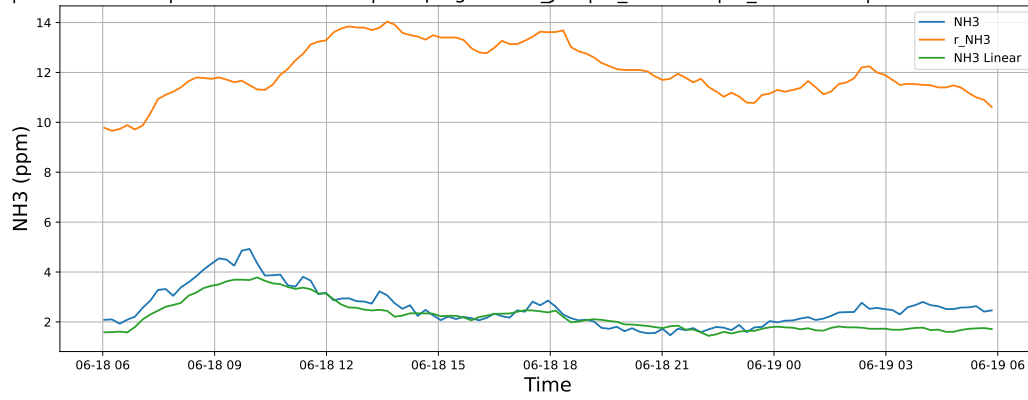
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



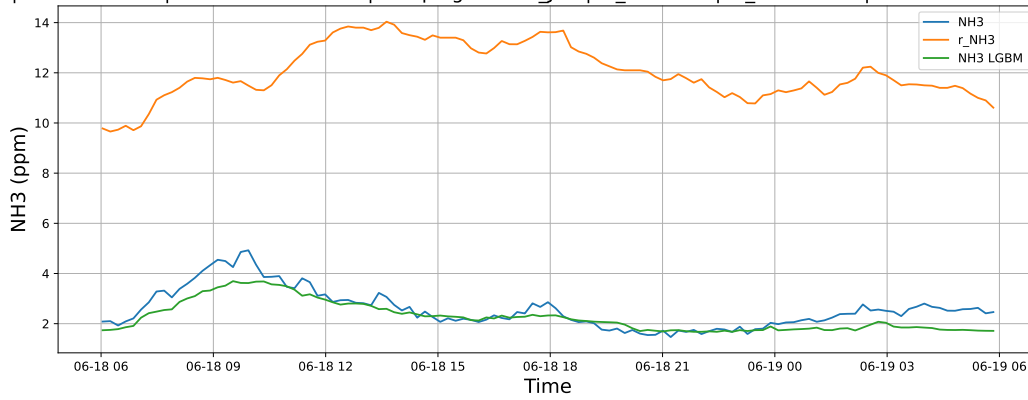
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

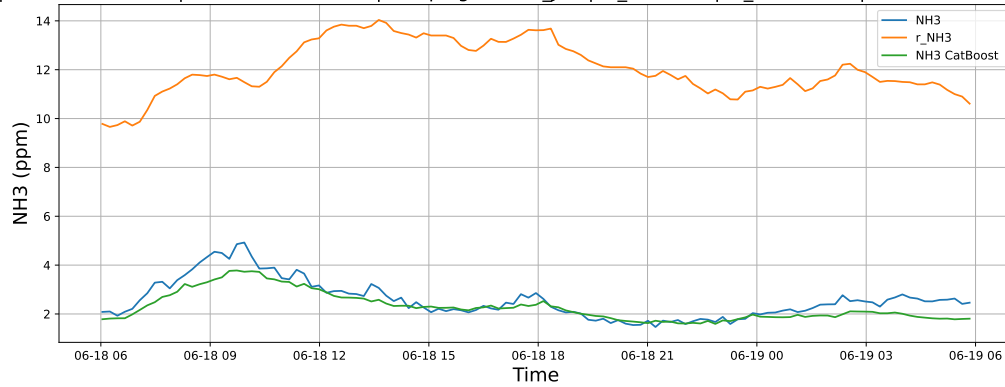
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

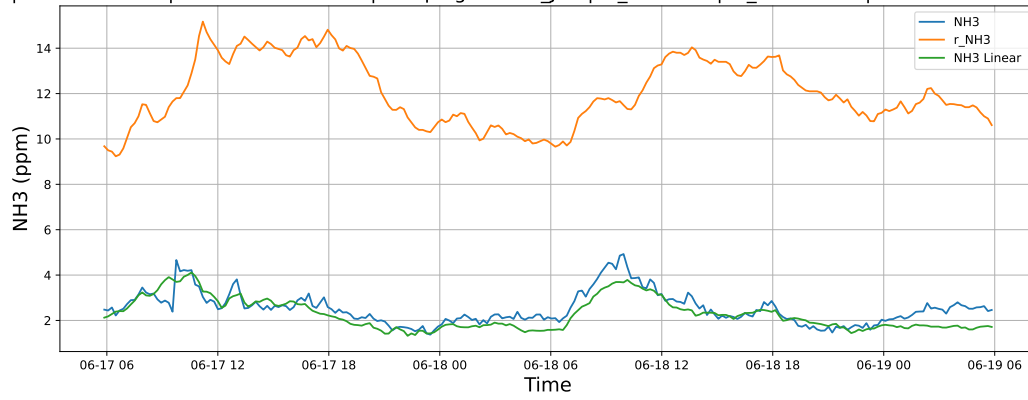


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

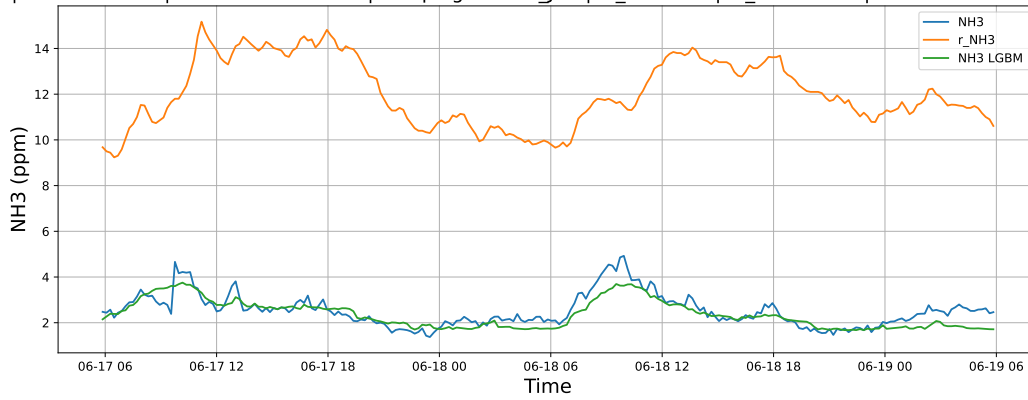


3.1.3.2 Time window : 48

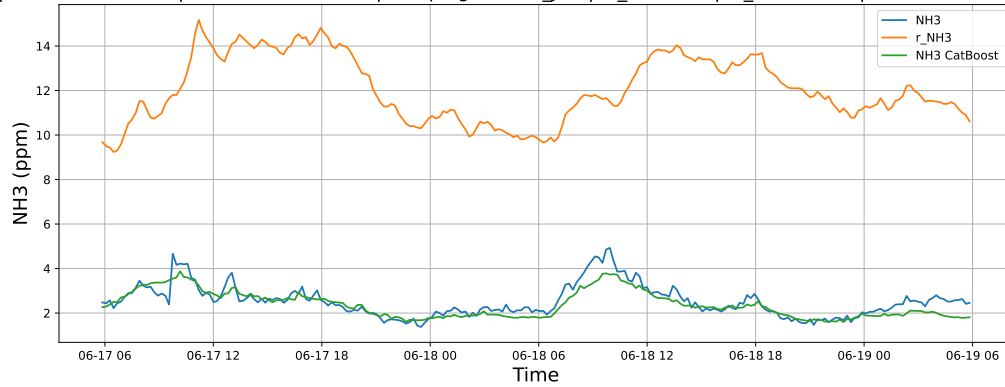
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

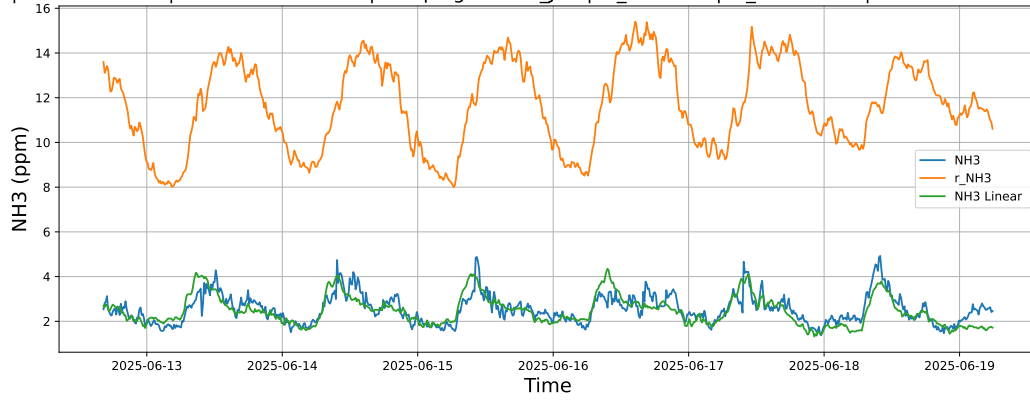


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

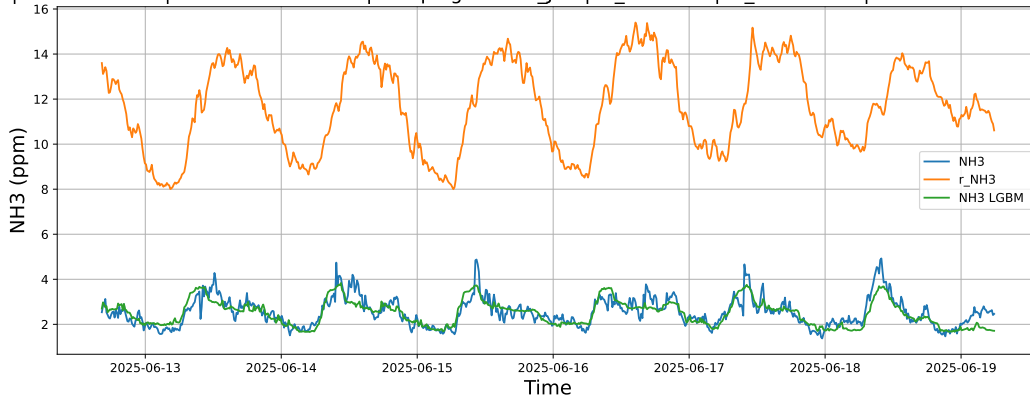


3.1.3.3 Time window : ALL

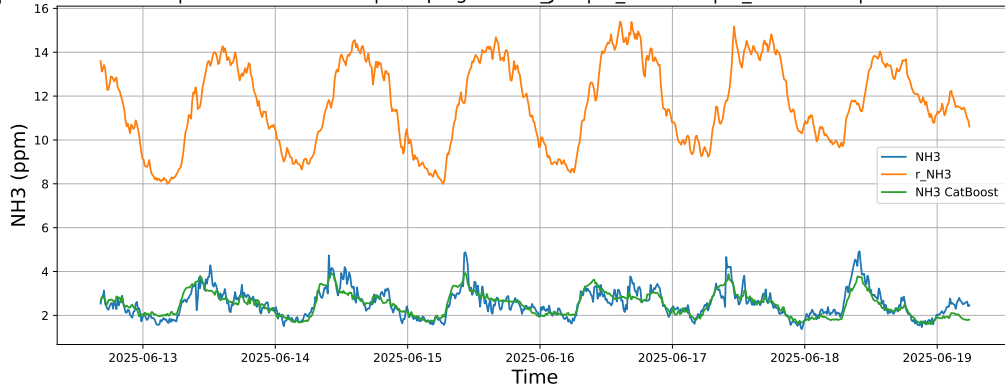
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

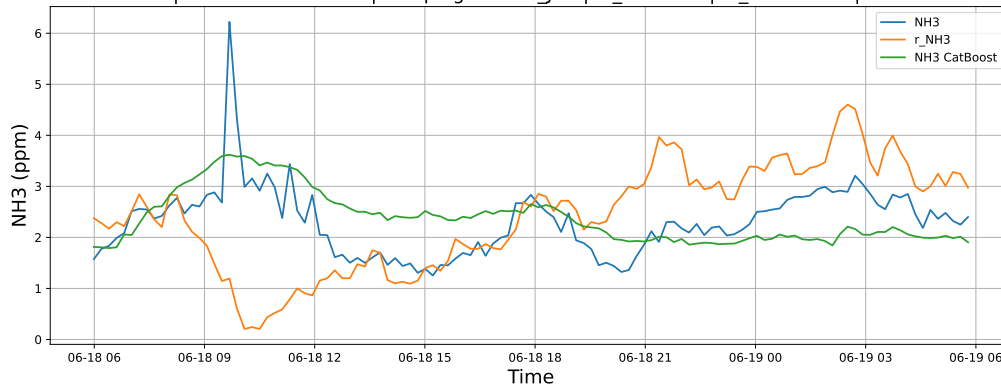
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

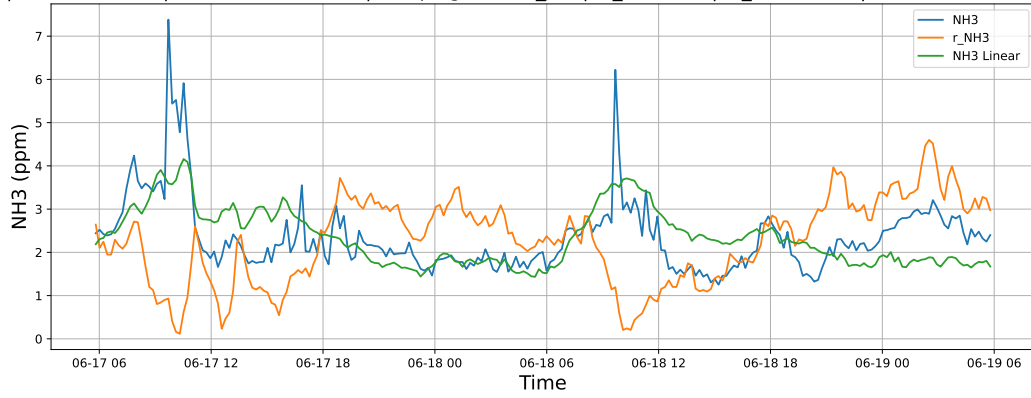


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

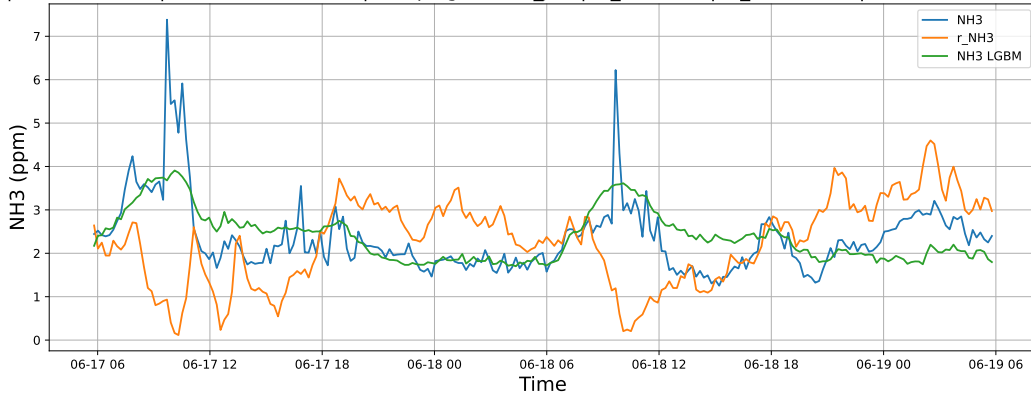


3.1.4.2 Time window : 48

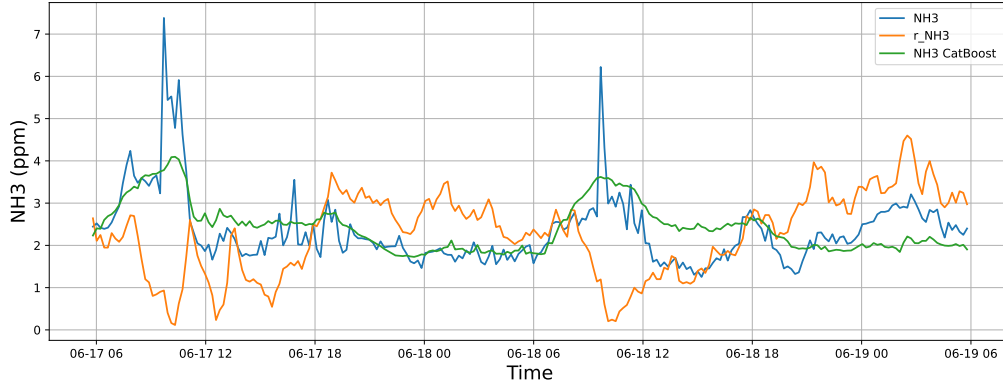
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

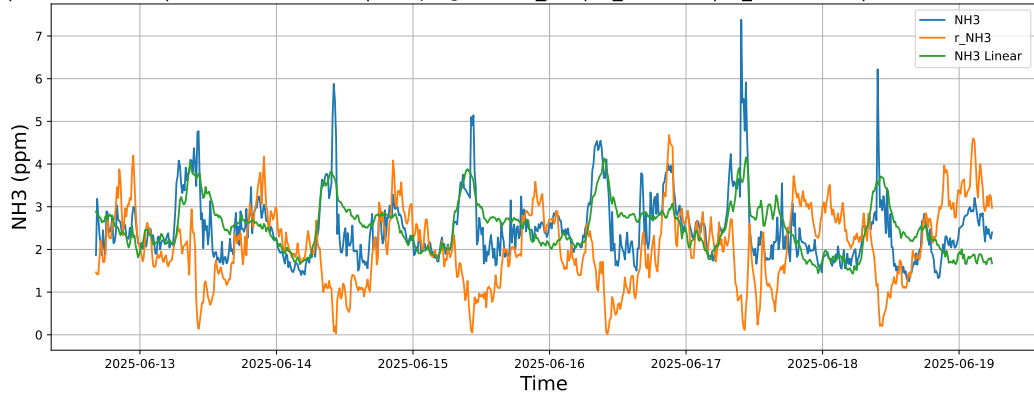


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

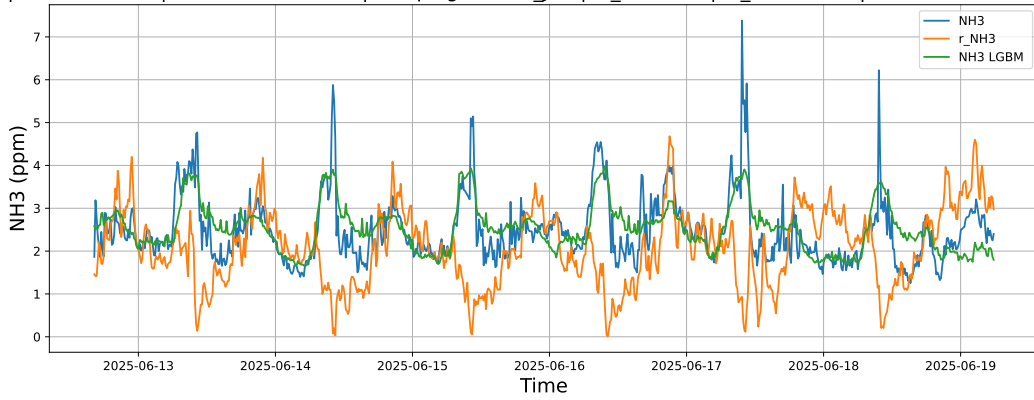


3.1.4.3 Time window : ALL

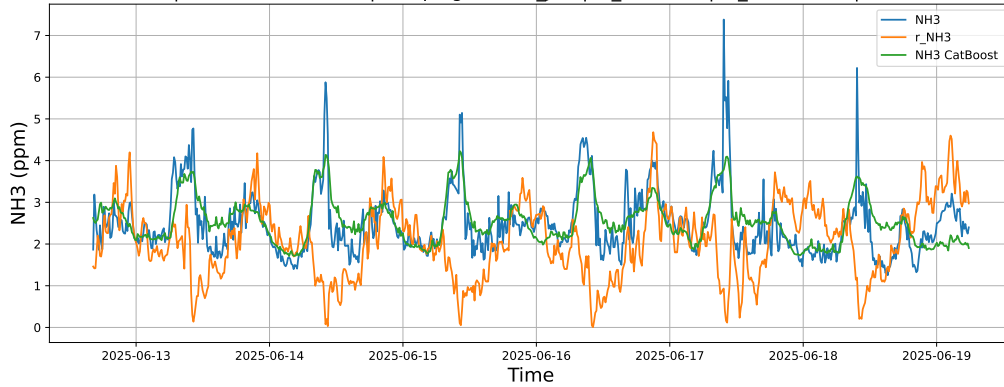
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



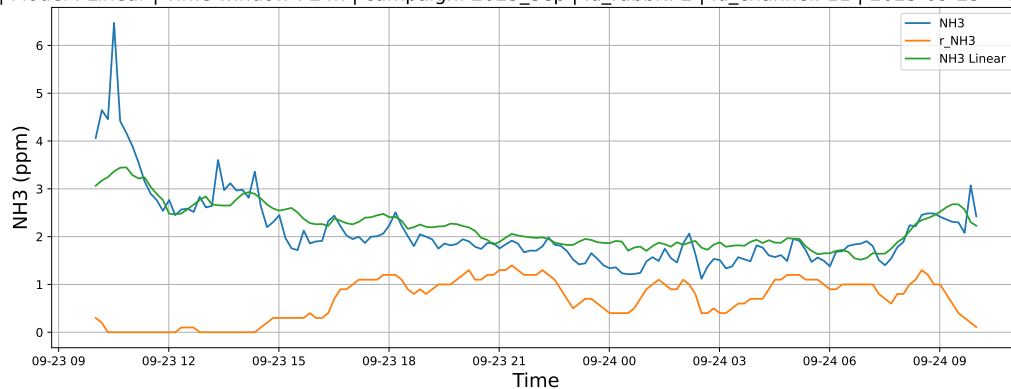
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



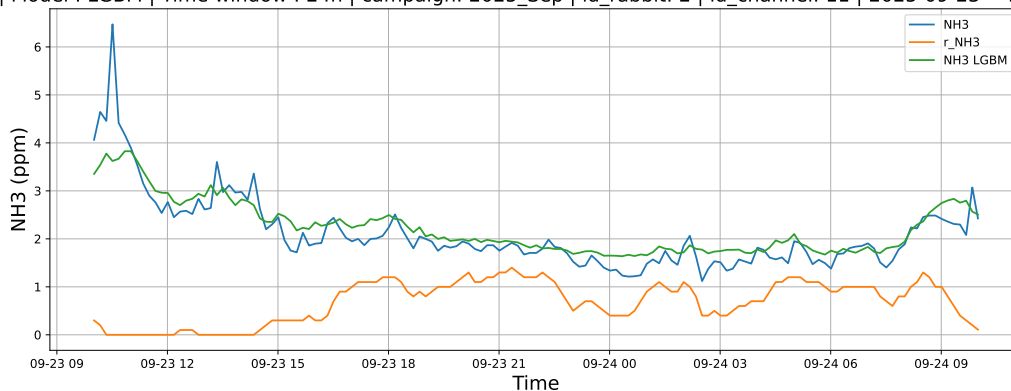
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

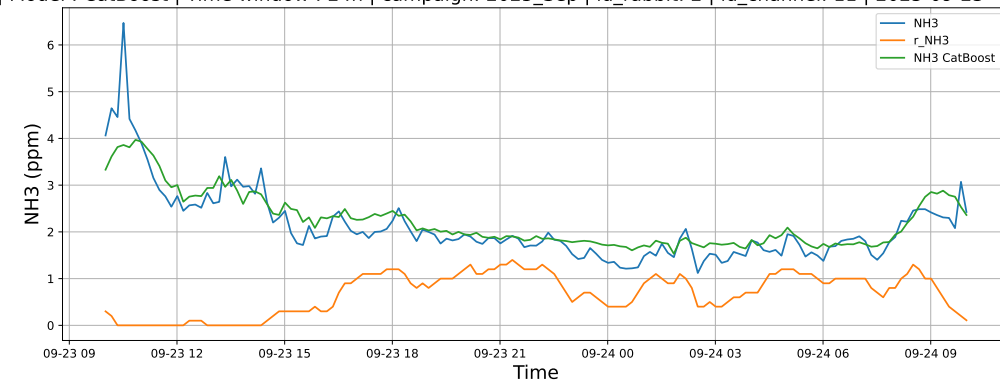
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

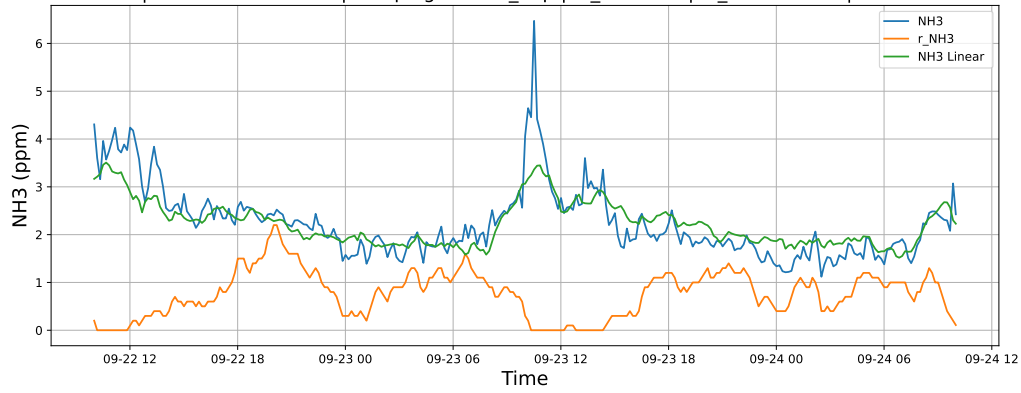


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

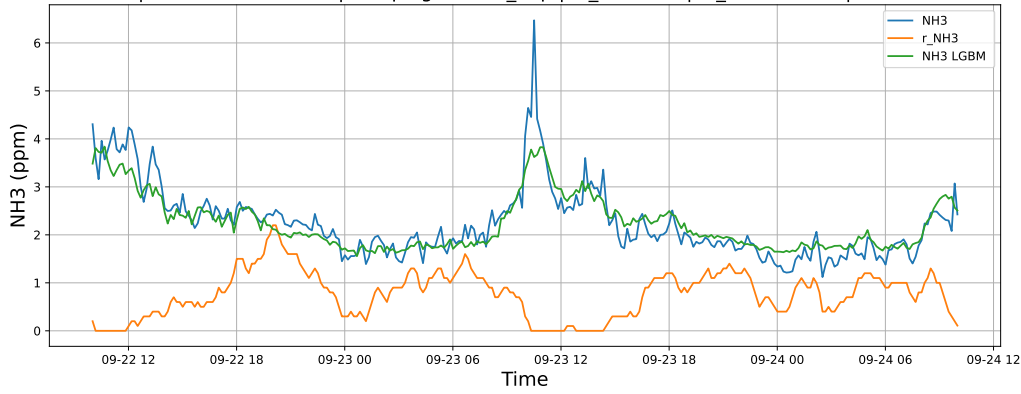


3.1.5.2 Time window : 48

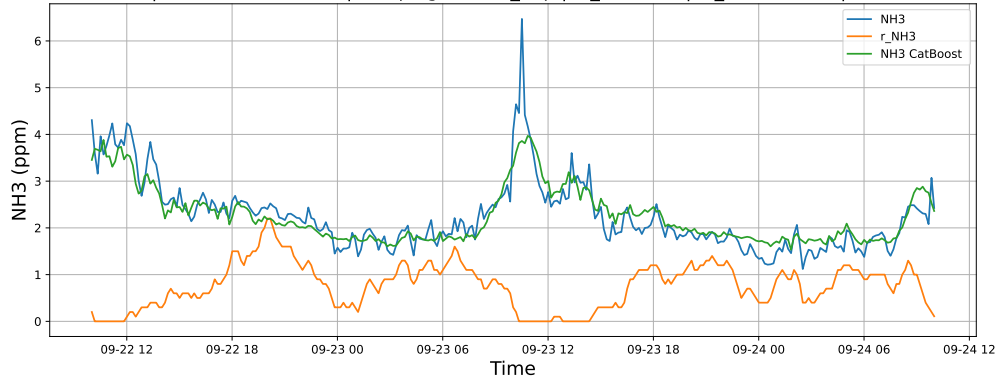
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

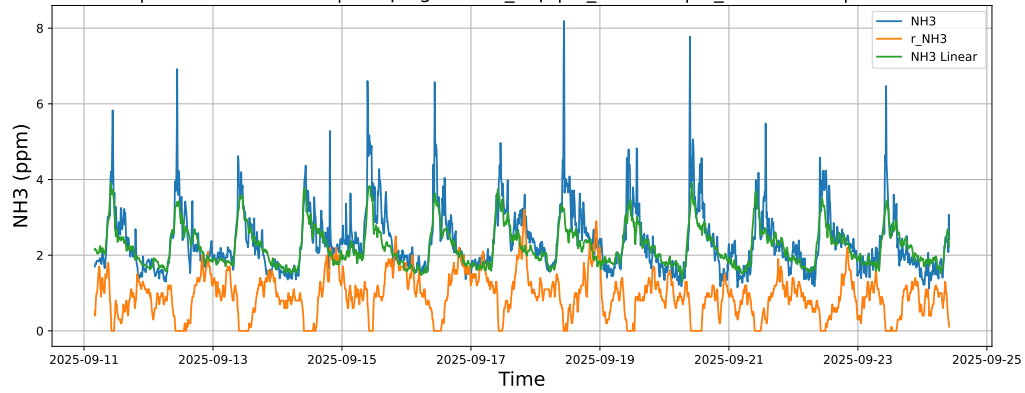


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

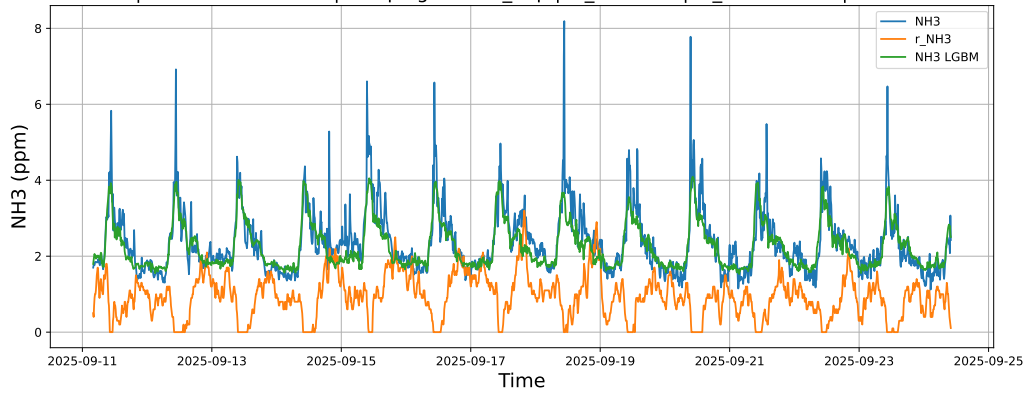


3.1.5.3 Time window : ALL

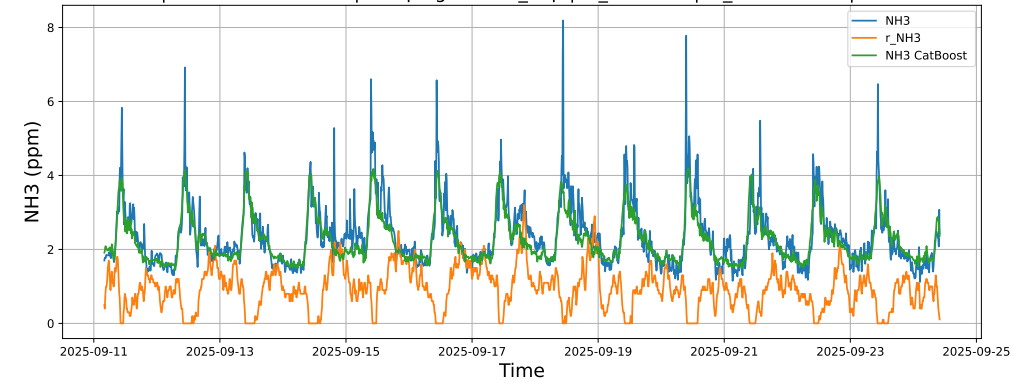
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



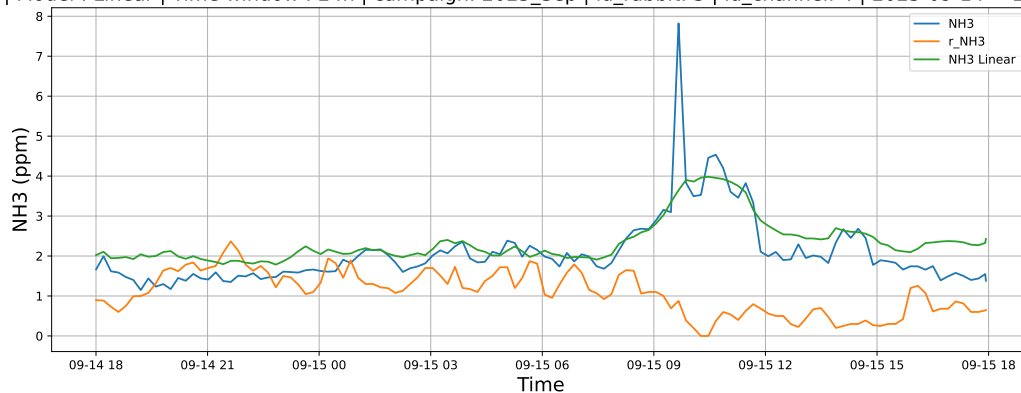
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



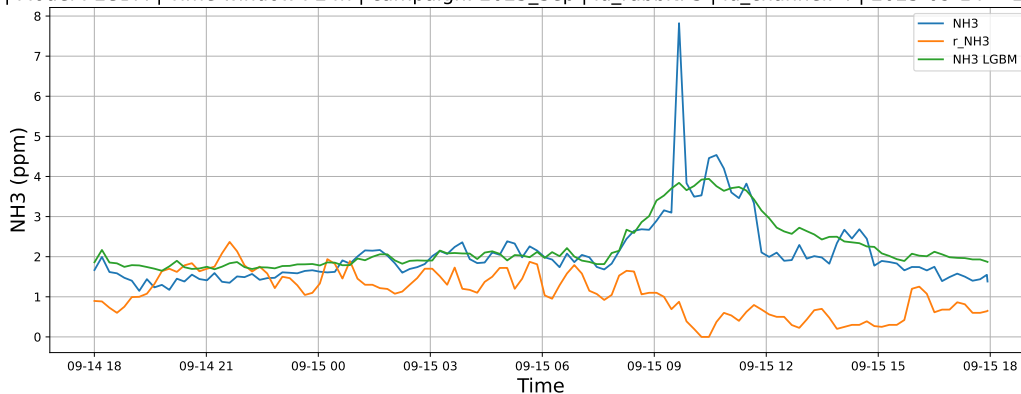
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

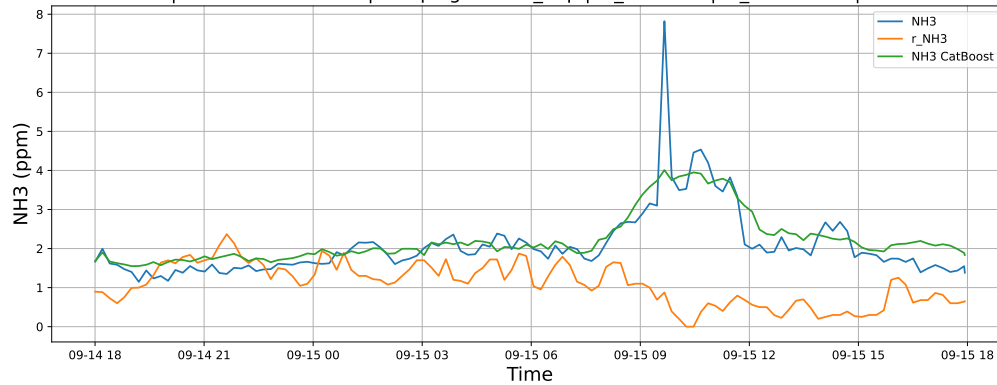
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

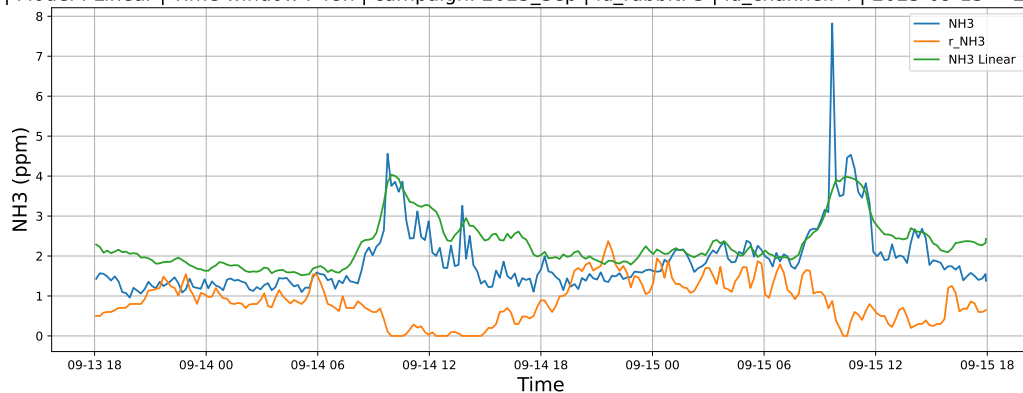


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

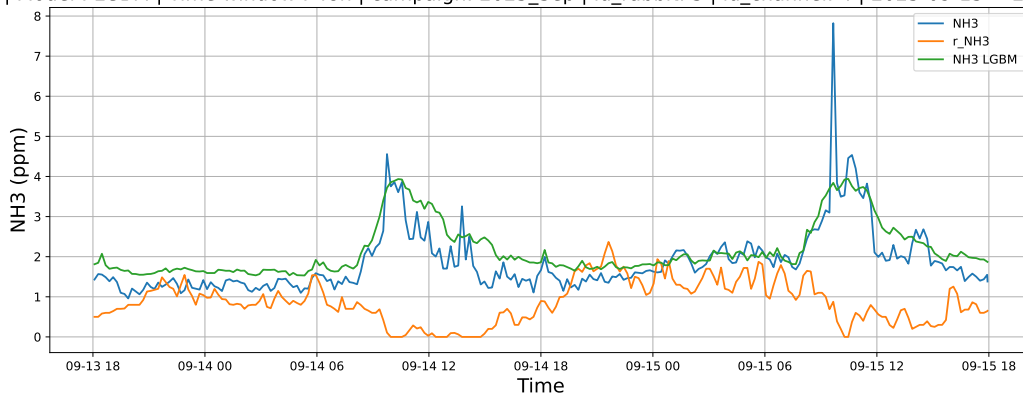


3.1.6.2 Time window : 48

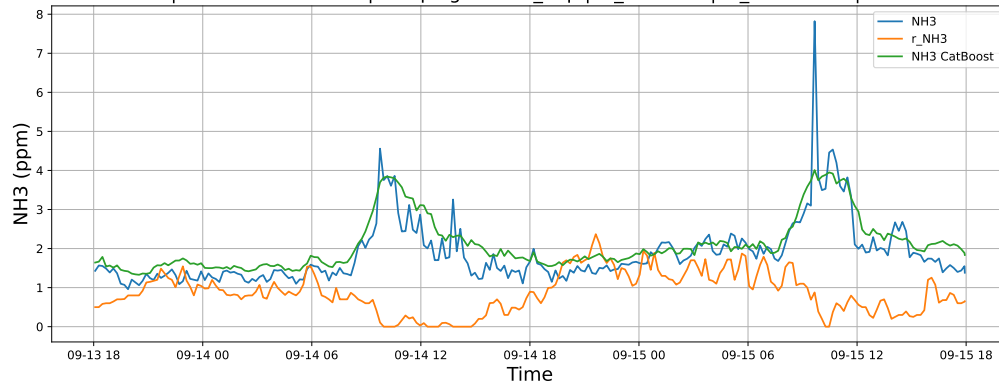
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

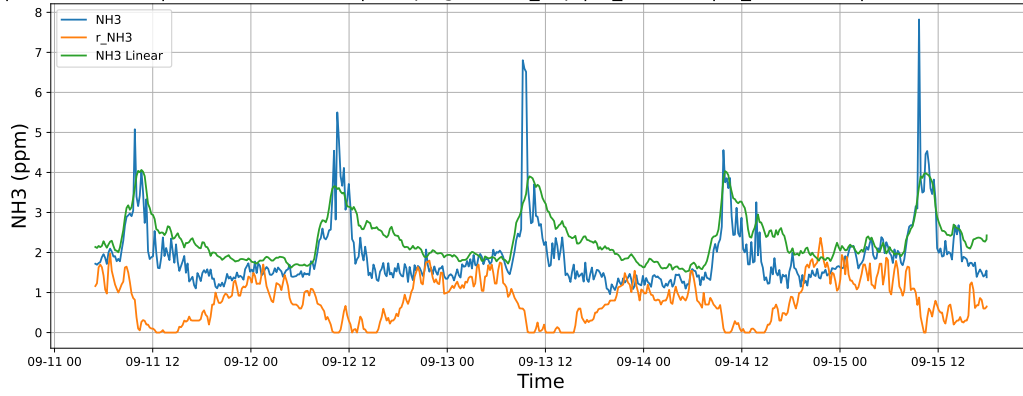


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

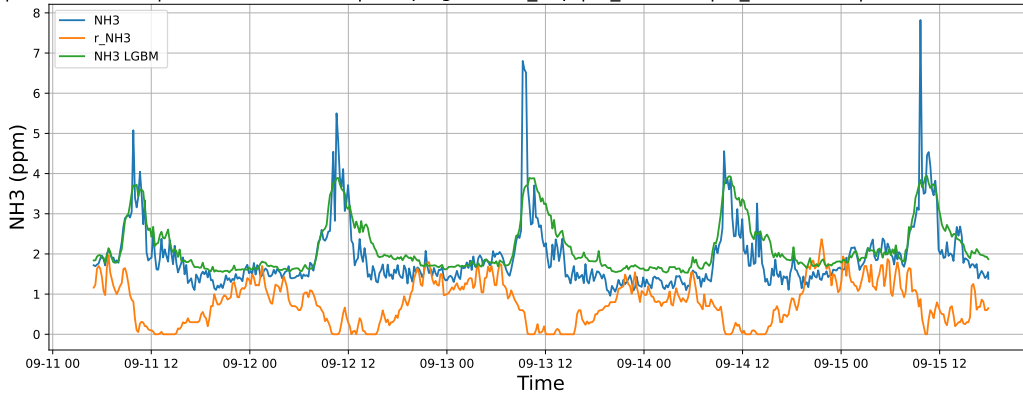


3.1.6.3 Time window : ALL

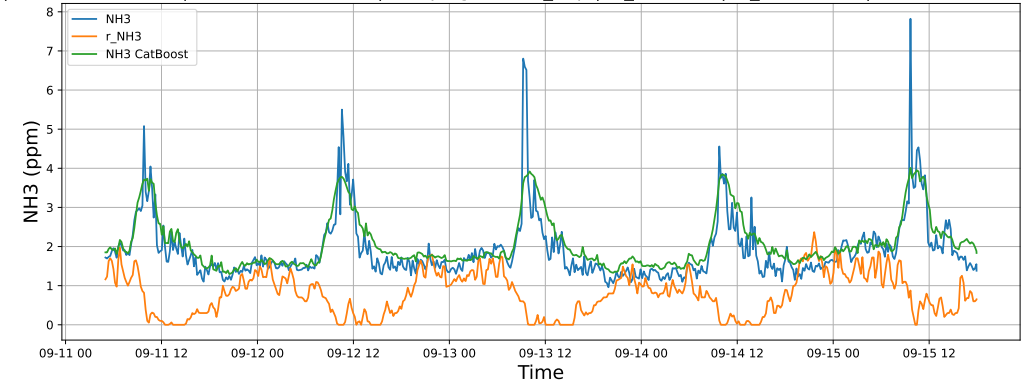
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



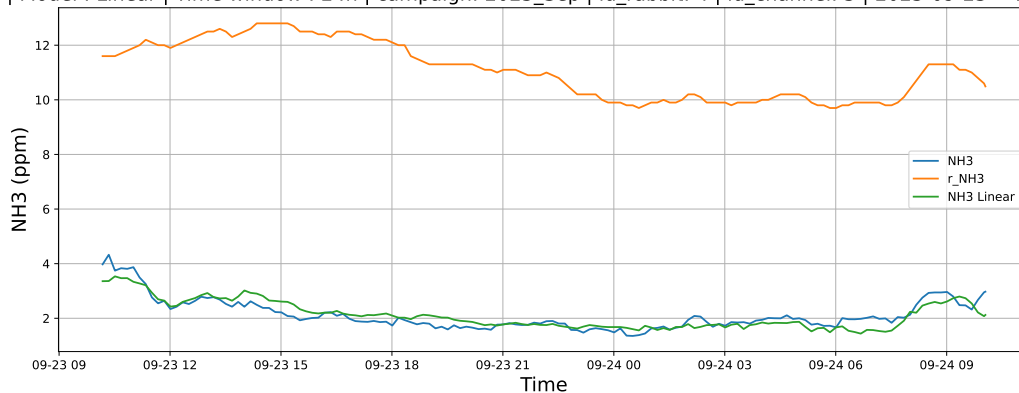
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



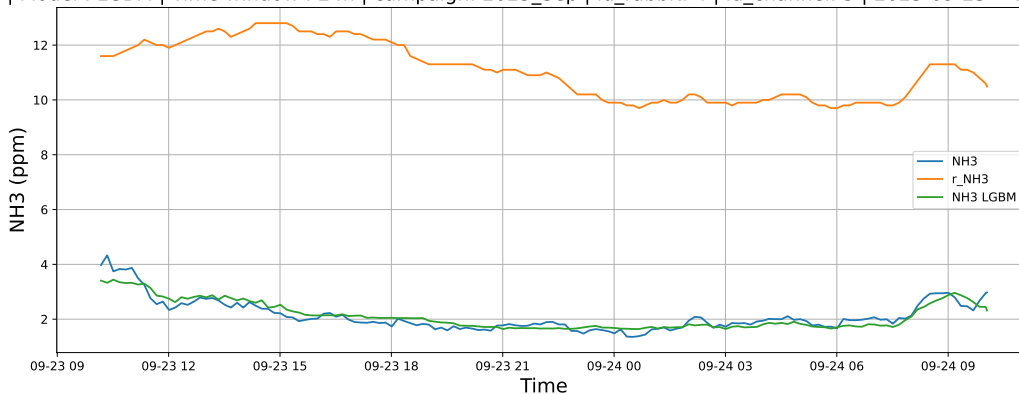
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

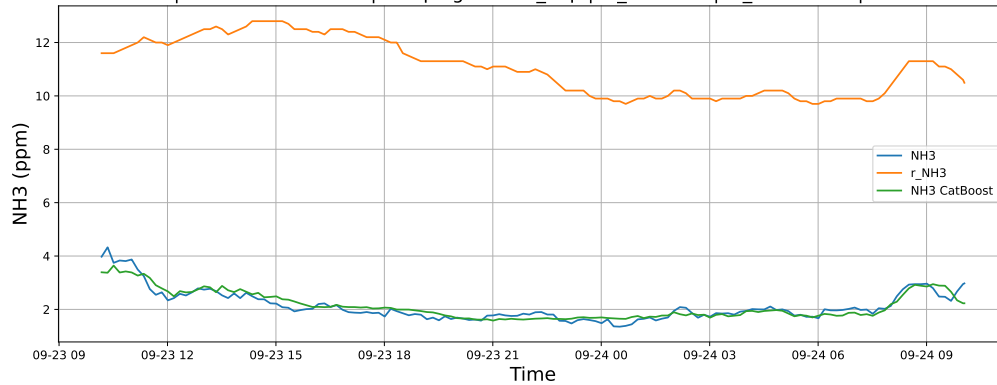
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

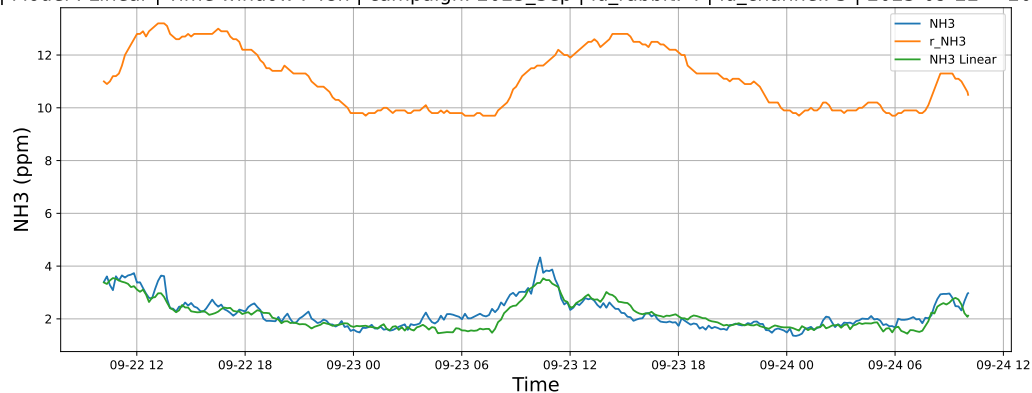


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

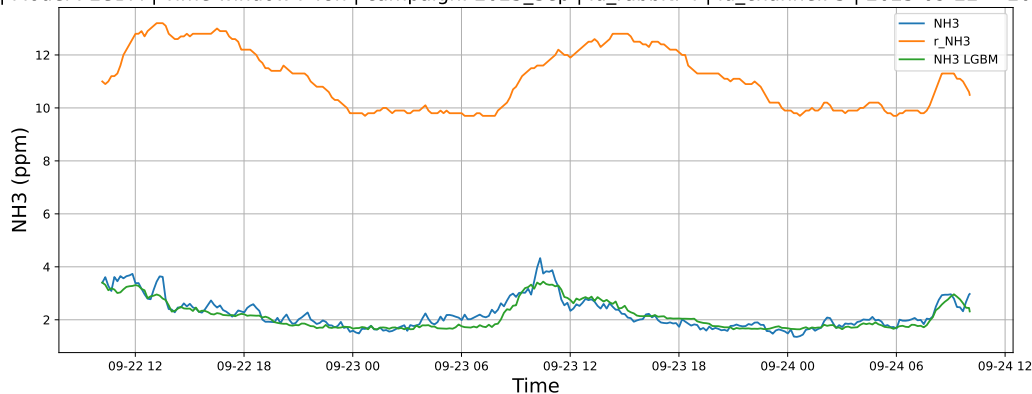


3.1.7.2 Time window : 48

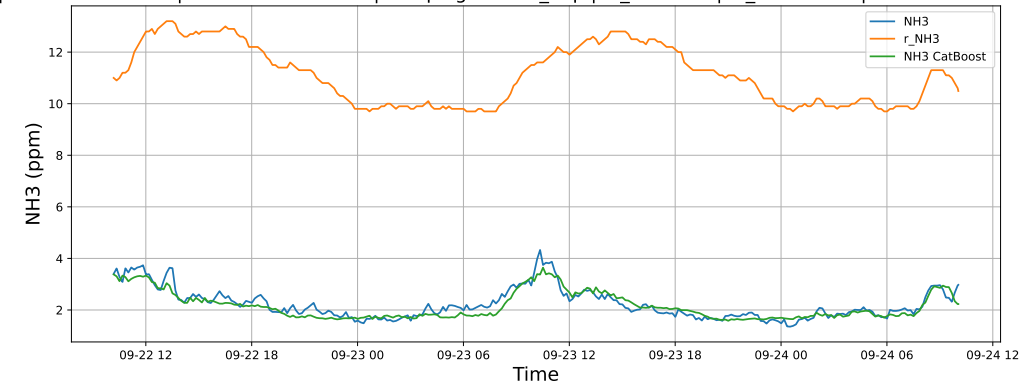
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

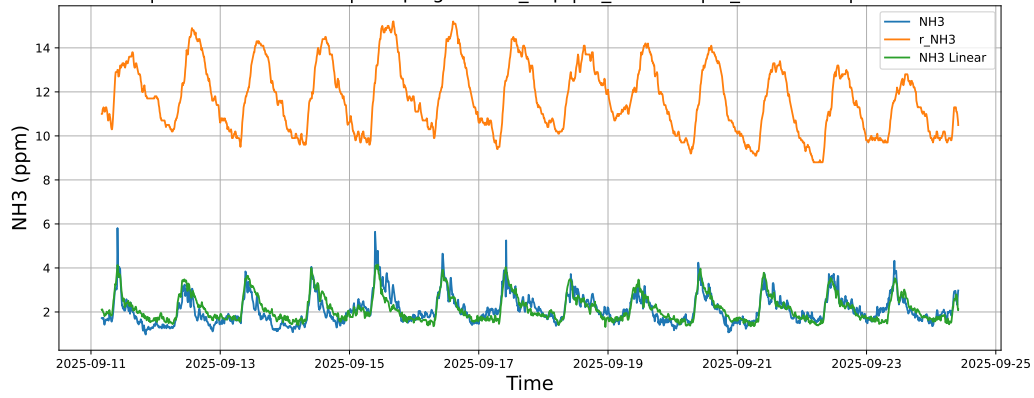


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

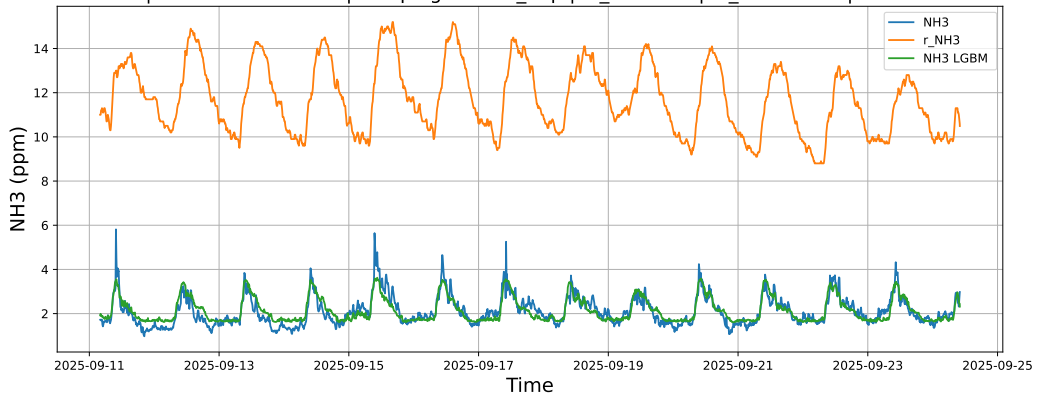


3.1.7.3 Time window : ALL

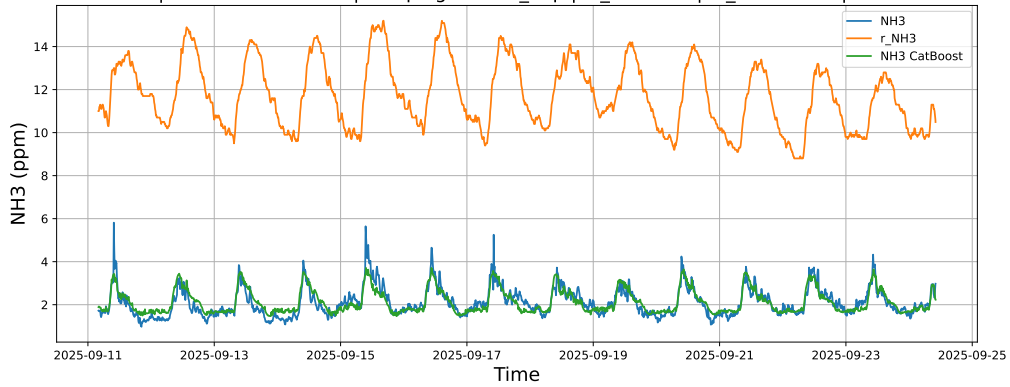
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



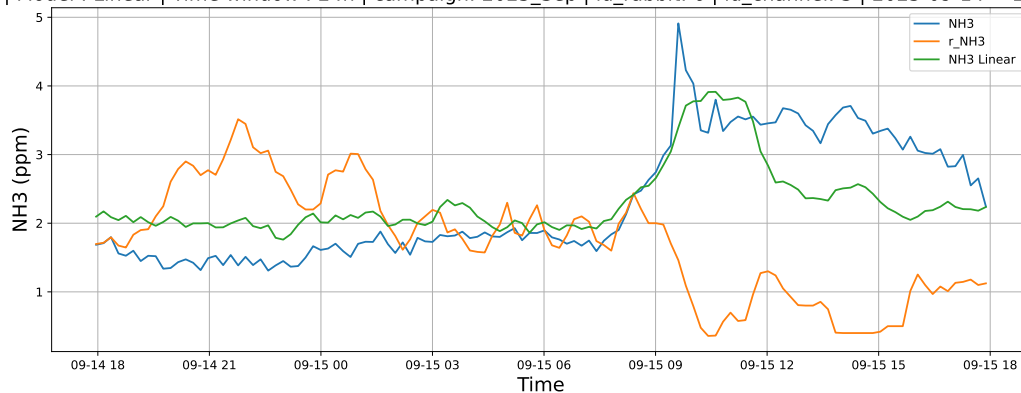
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



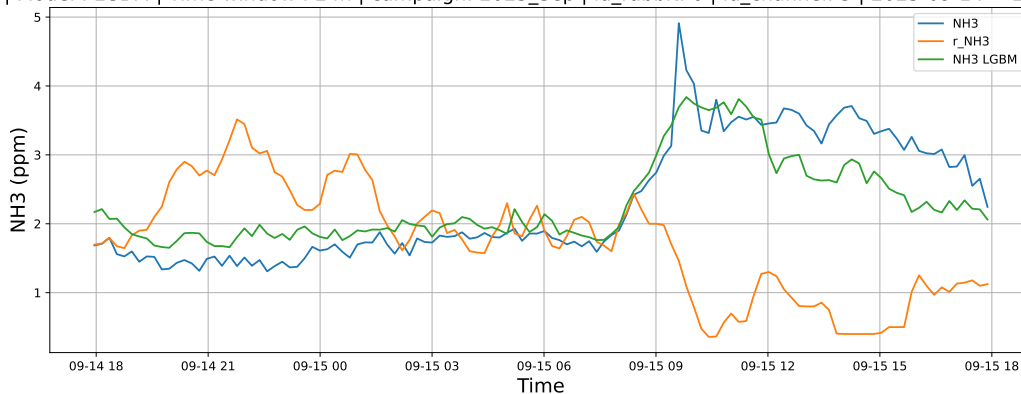
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

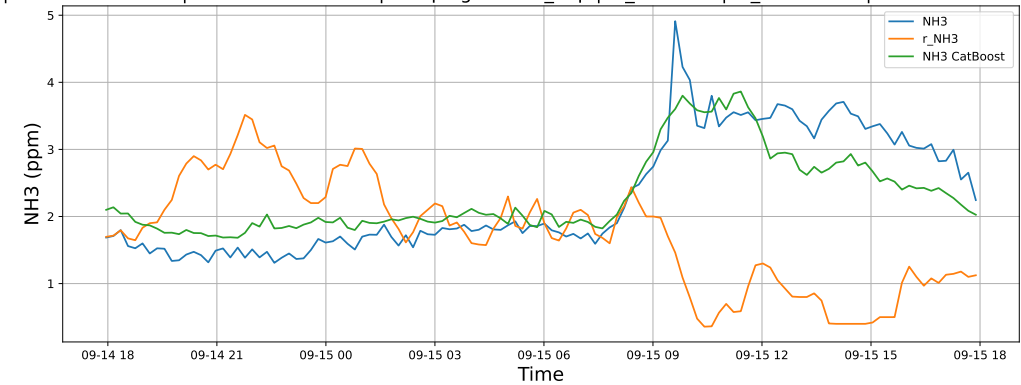
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

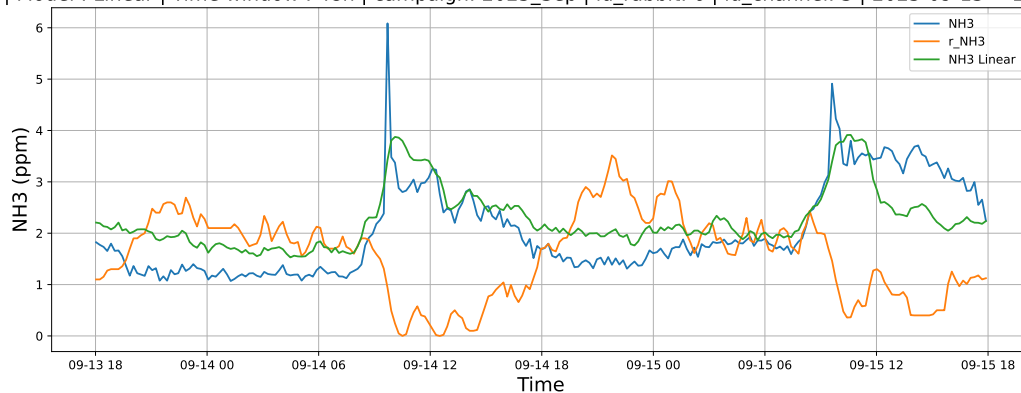


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

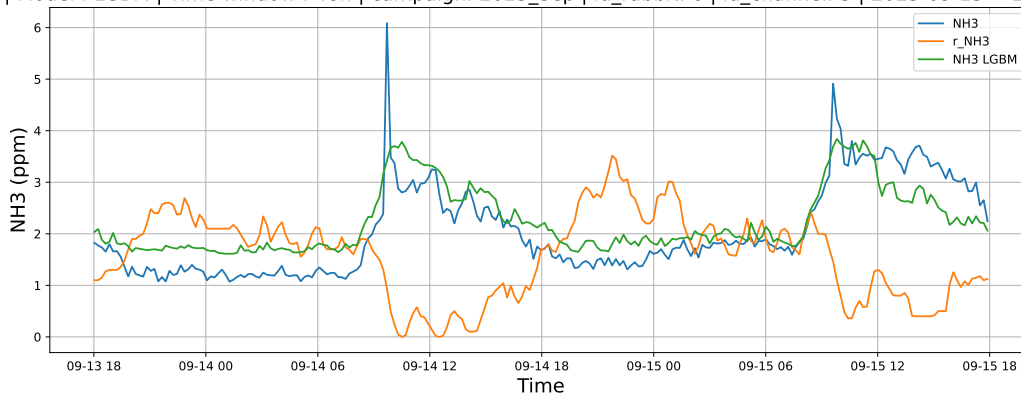


3.1.8.2 Time window : 48

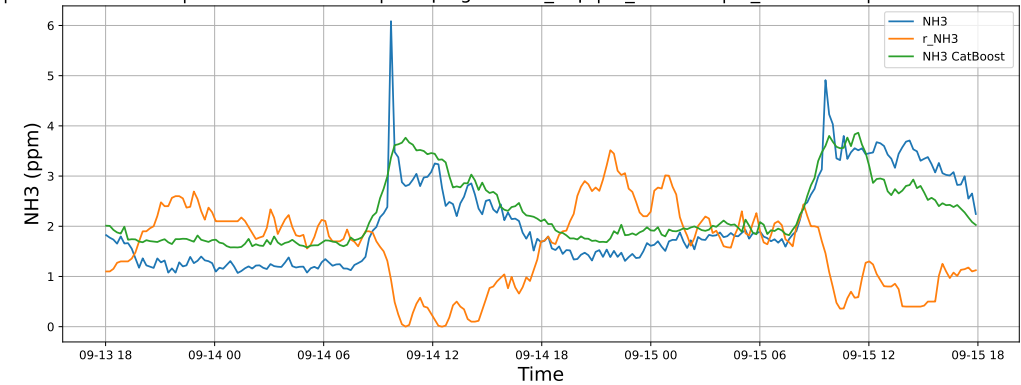
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

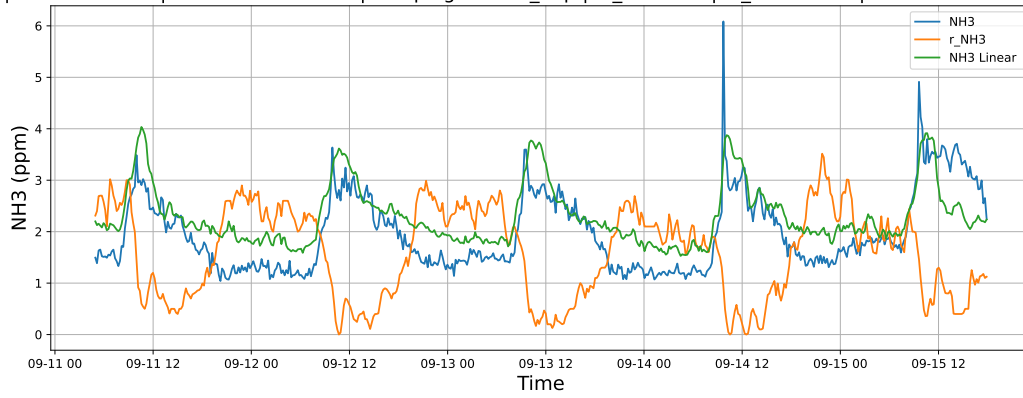


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

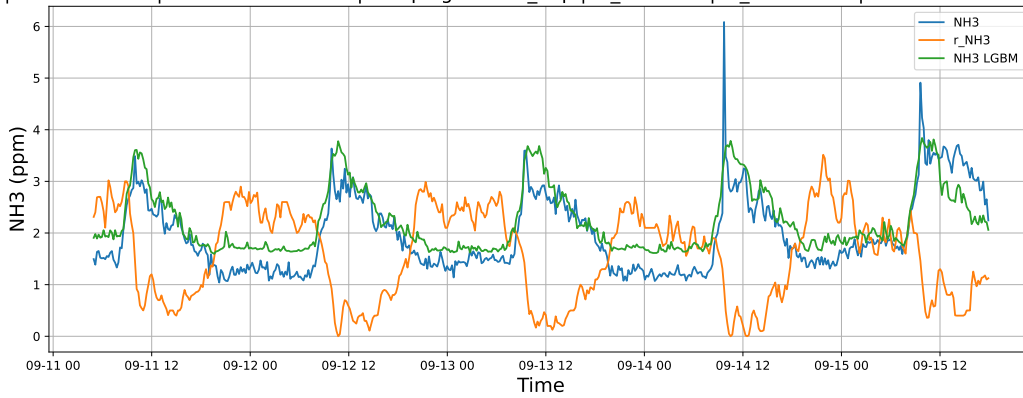


3.1.8.3 Time window : ALL

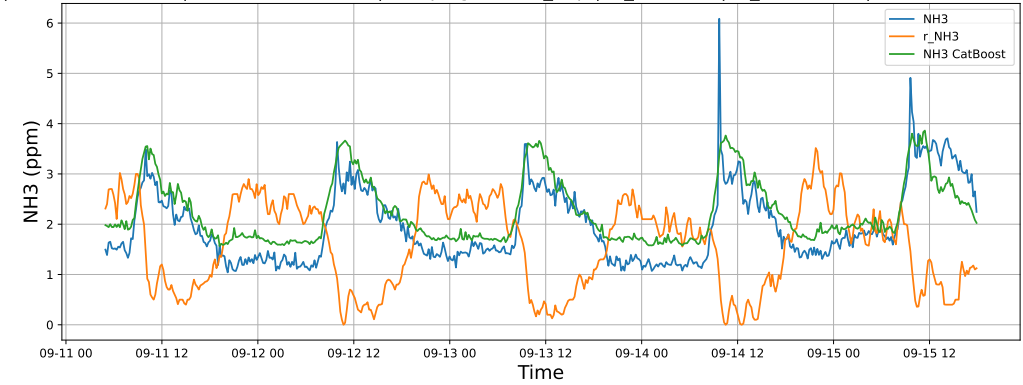
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

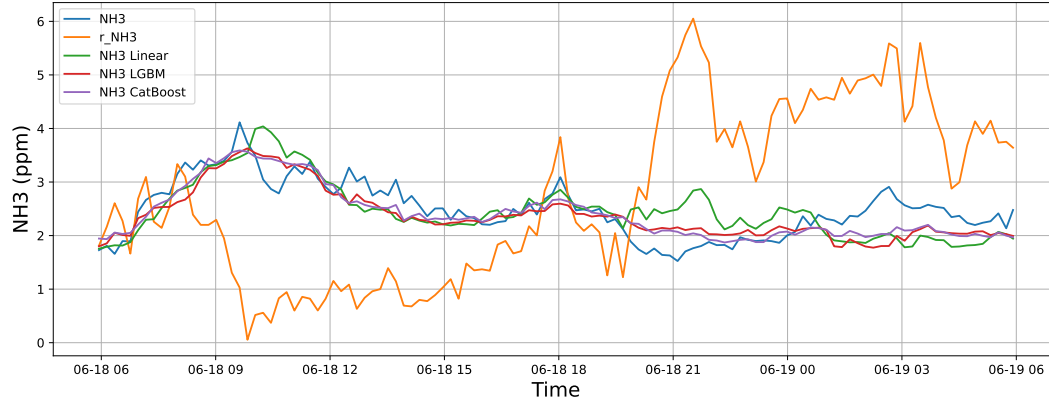


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

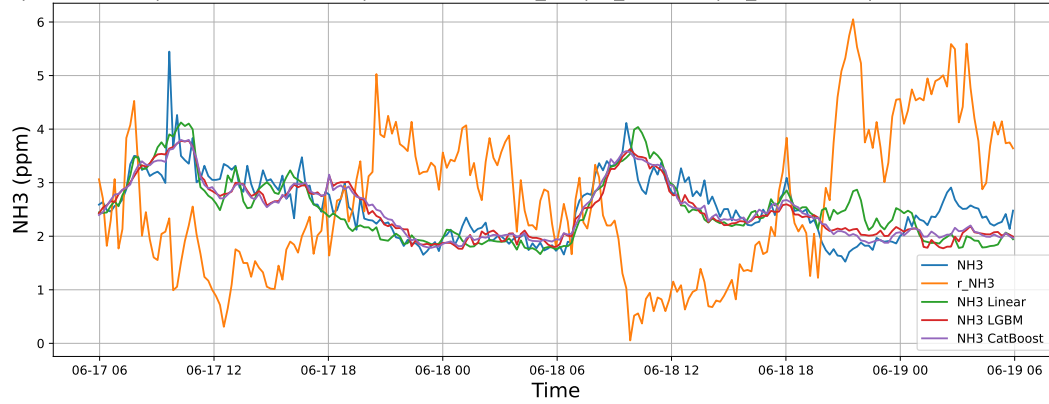
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



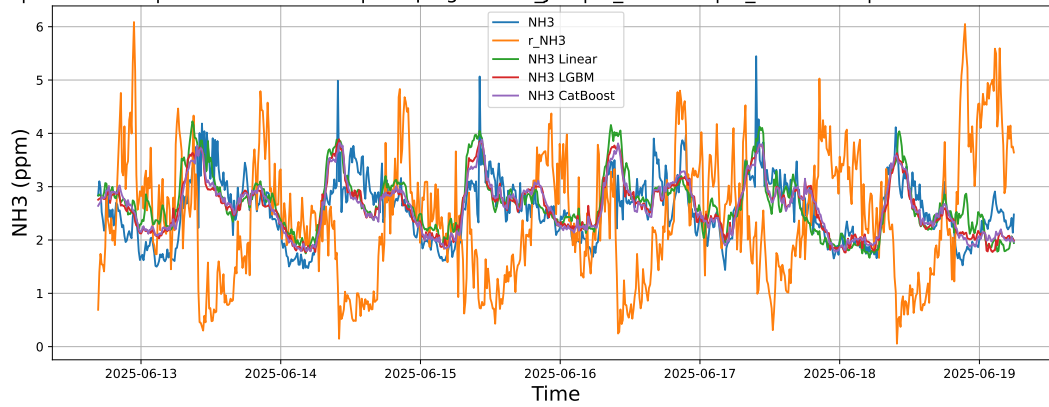
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

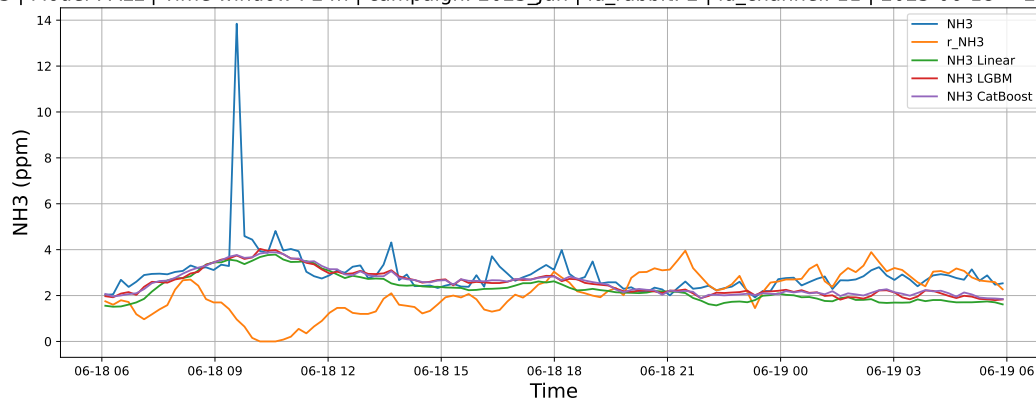
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

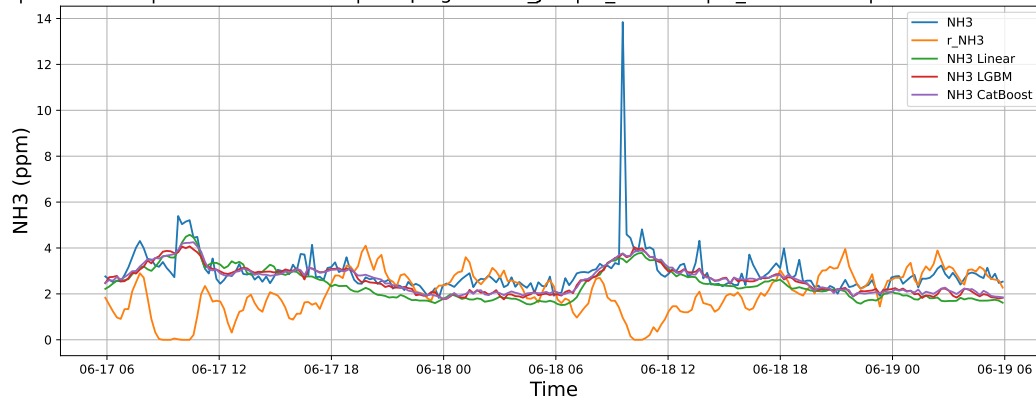
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



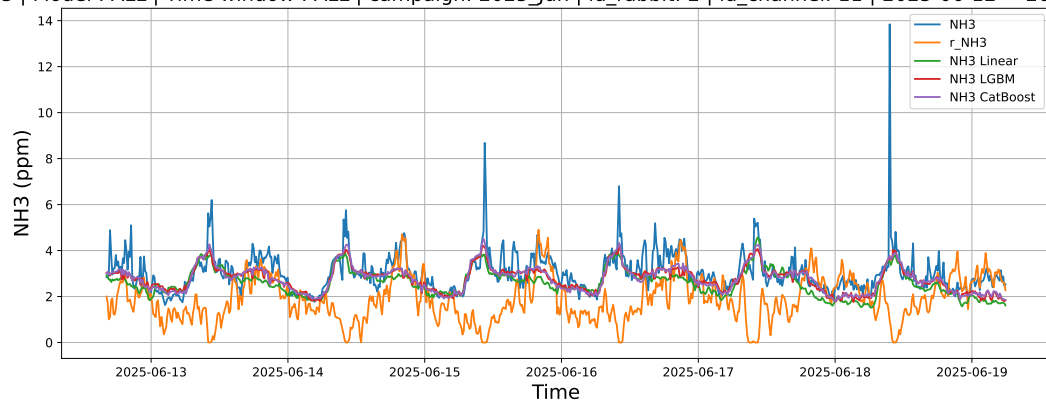
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

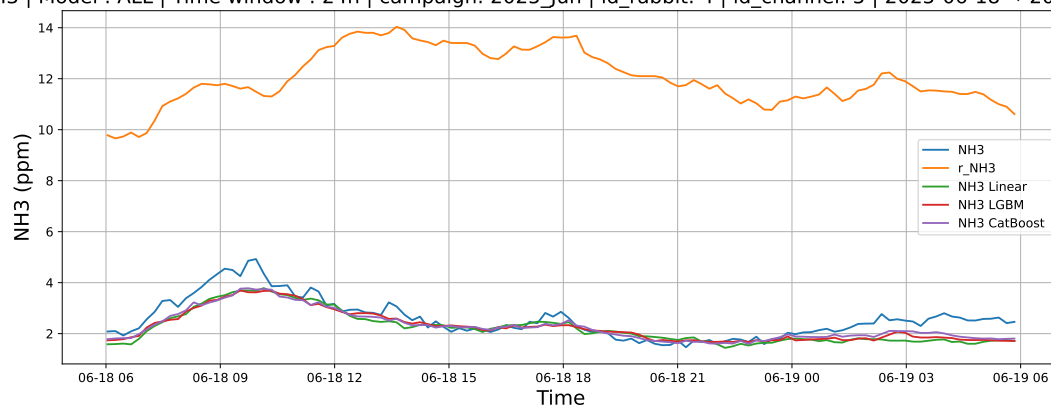
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

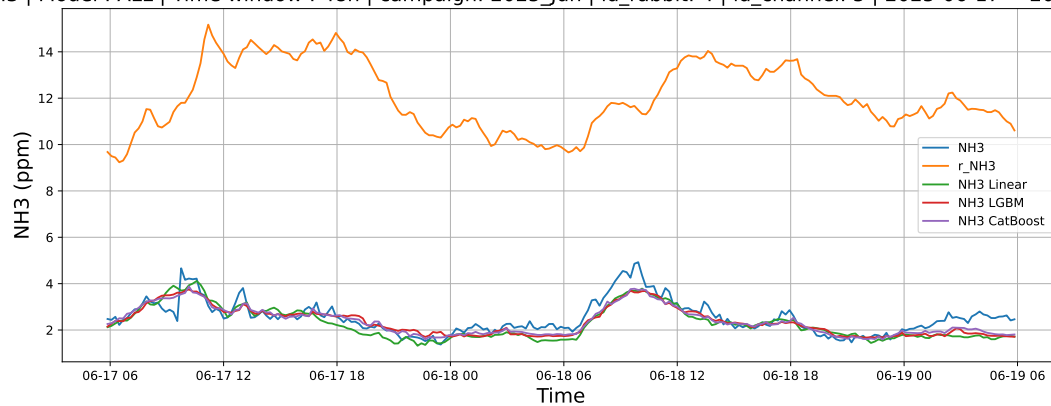
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



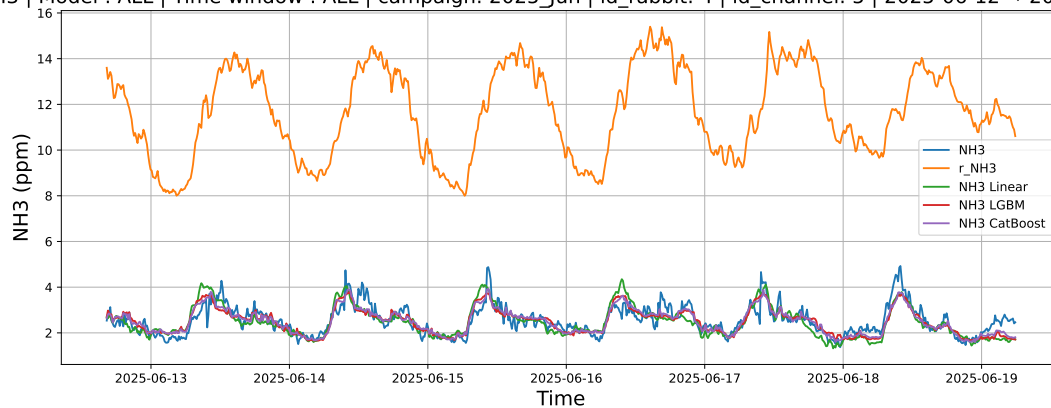
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

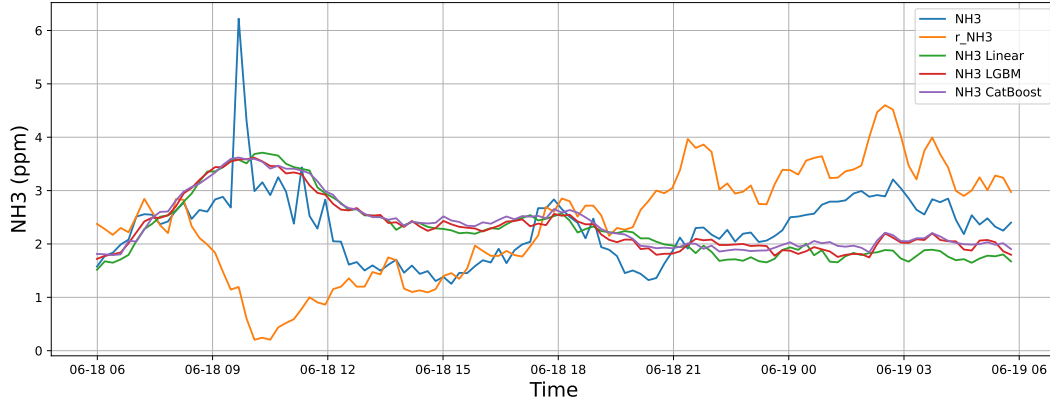
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

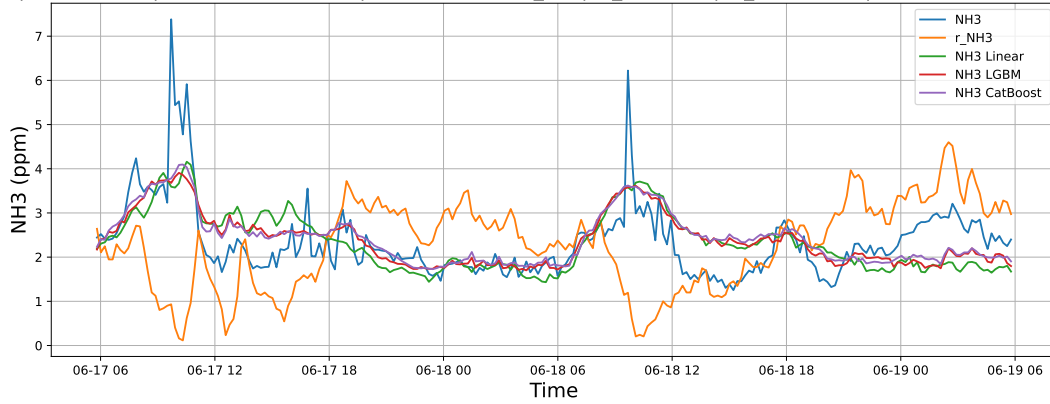
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



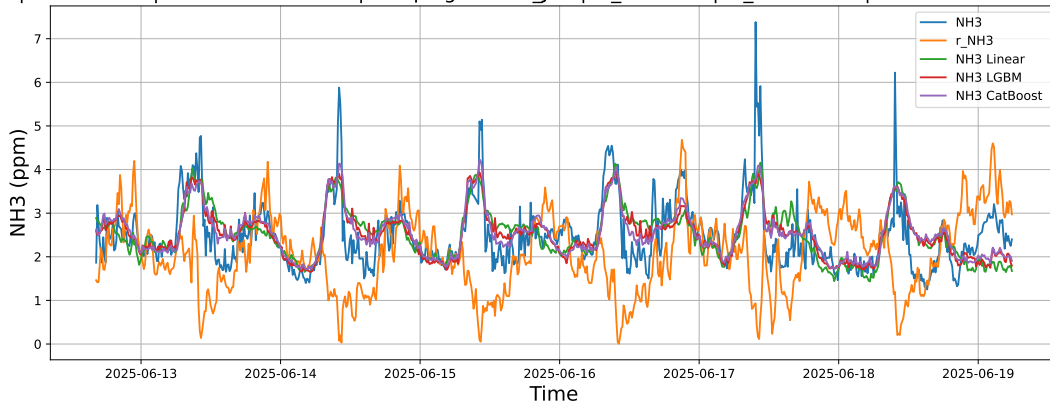
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

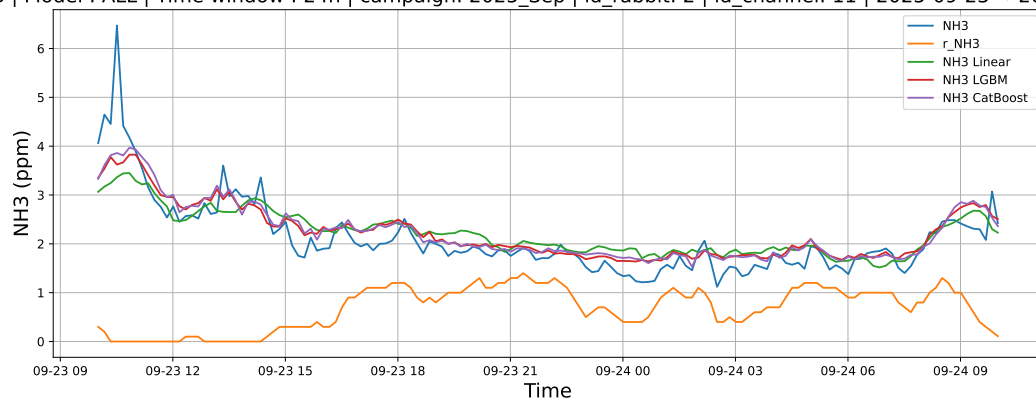
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

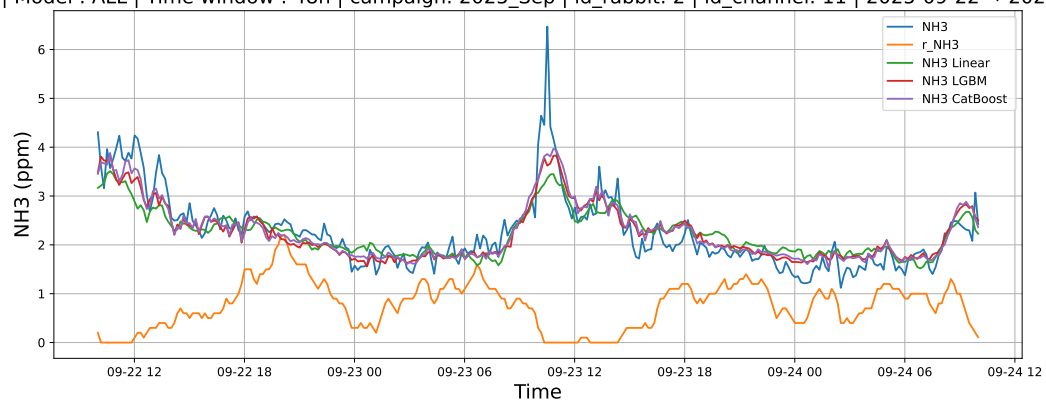
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



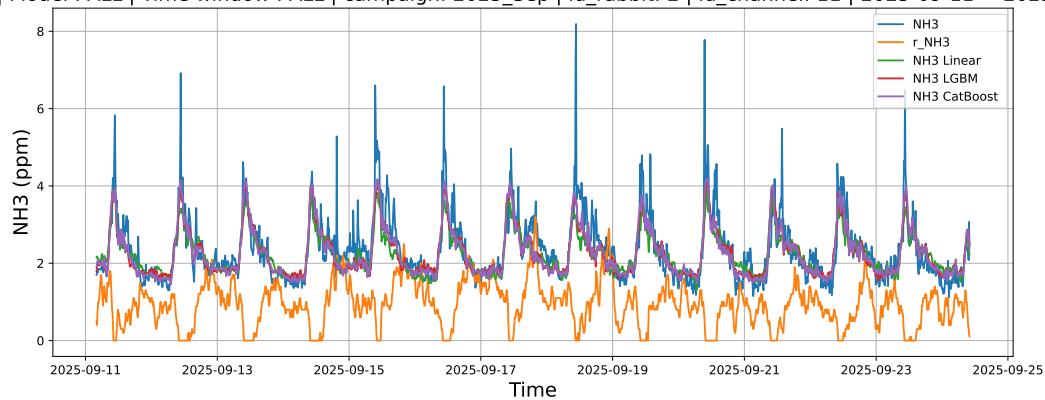
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

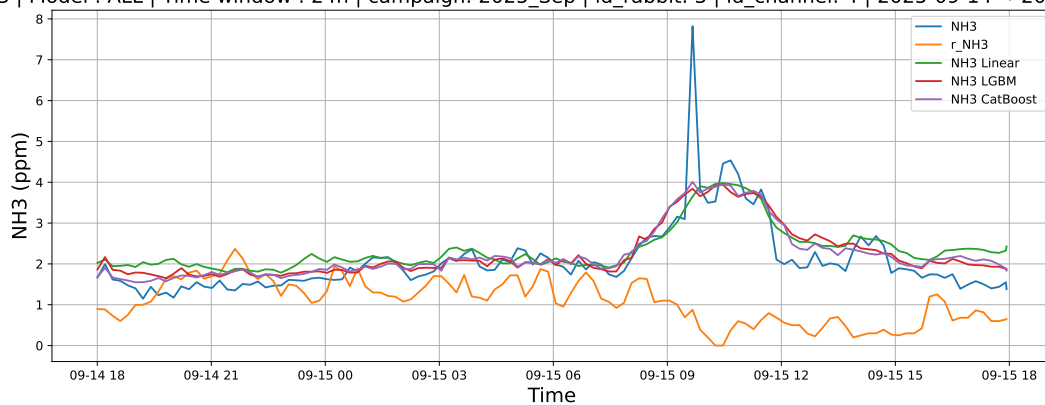
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

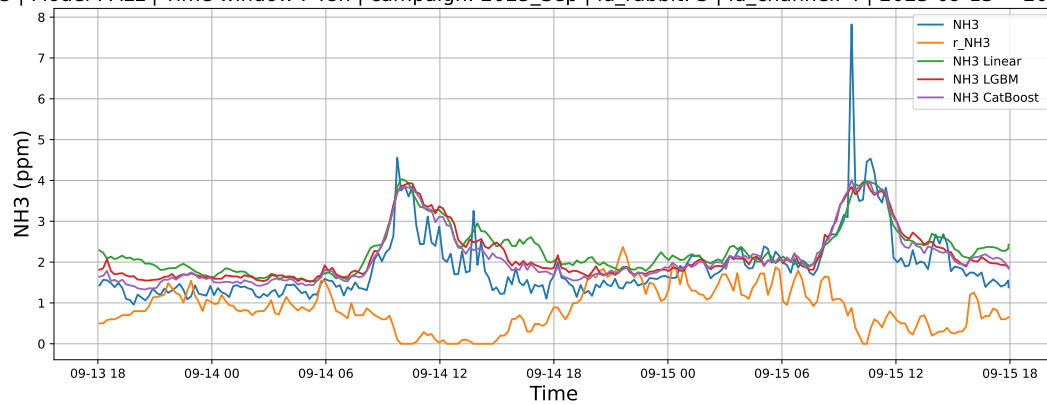
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



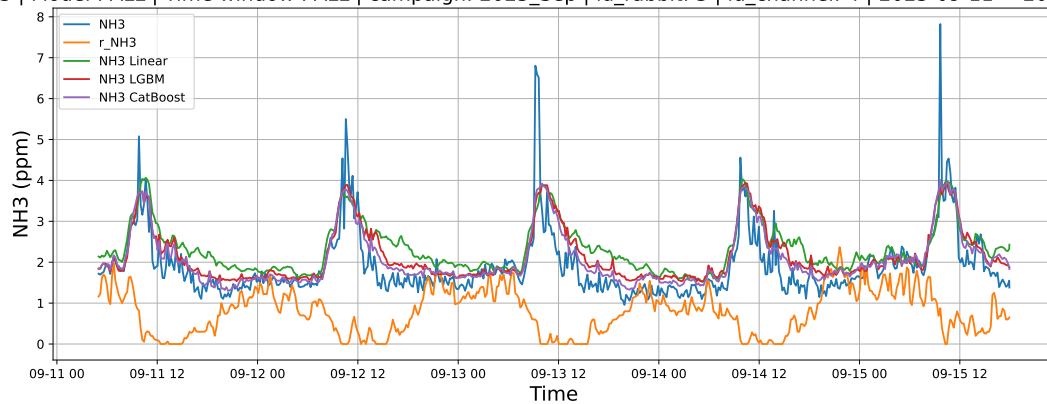
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

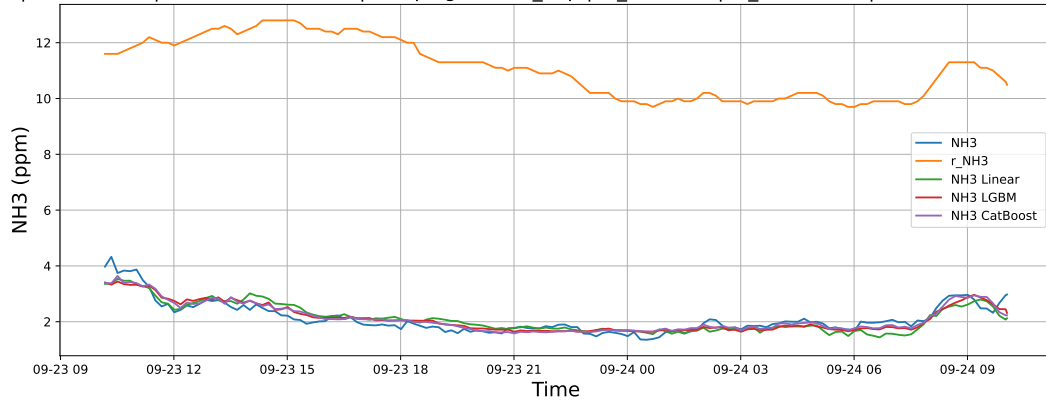
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

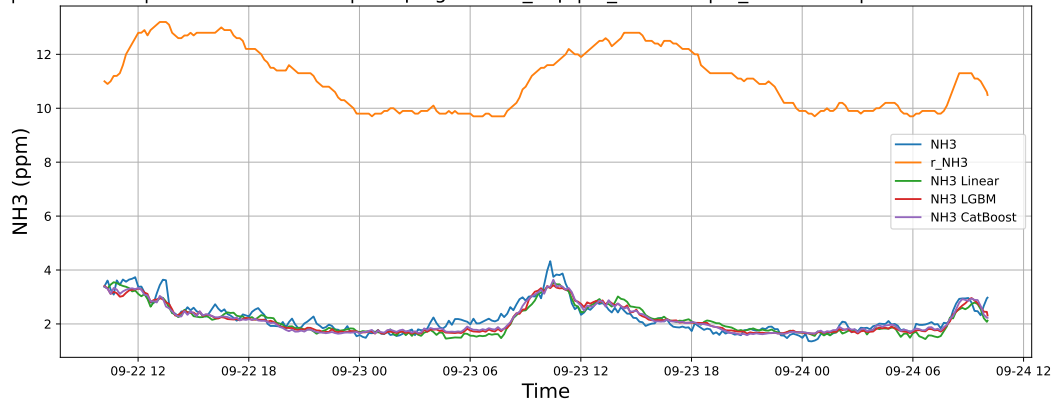
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



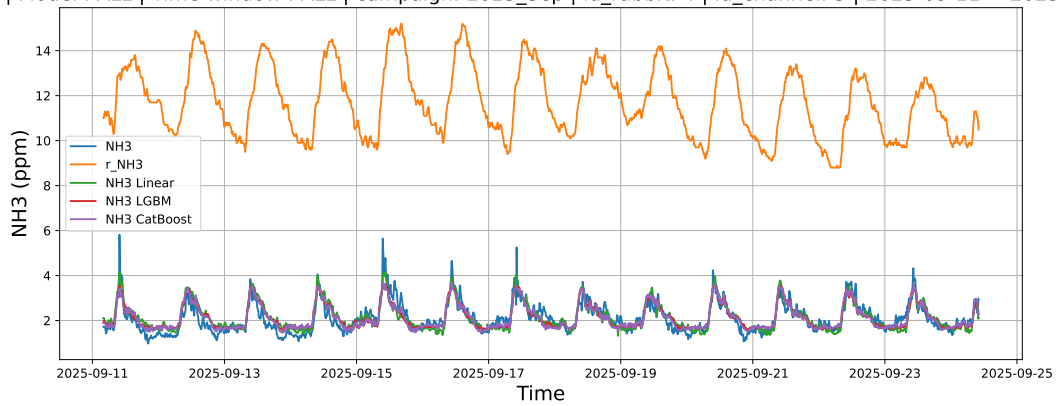
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

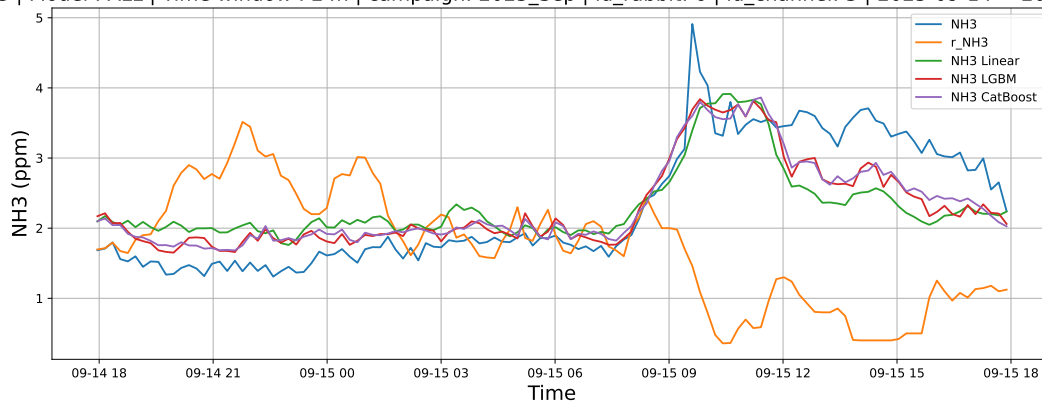
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

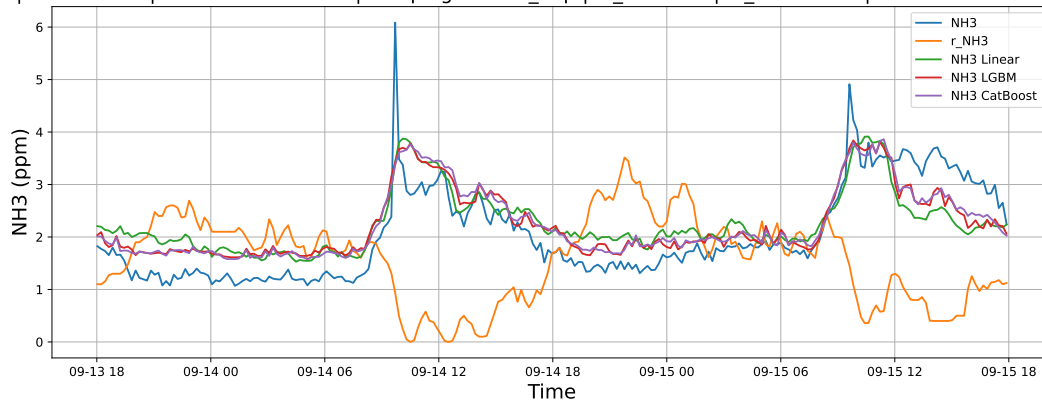
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

